Quality Screening through Market Access in Public Infrastructure: Evidence from Rwanda

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Abstract

Governments often cannot contract on the quality of what they procure. Screening may improve performance by restricting which firms can compete, but can also weaken competition and raise prices. In repeated procurement markets, governments can screen firms based on past performance. This paper studies whether screening on past performance can mitigate the price–quality tradeoff in Rwanda's public procurement of infrastructure, focusing on a key measure of quality: project completion time. I link the near-universe of electronic procurement records for 2016–2024 to records of government payments that trace contract execution. I estimate a dynamic auction model in which low- and high-quality firms compete in first-price auctions and completed work determines access to larger contracts. High-quality firms complete projects faster and upgrade sooner, so they value the future more. Despite having higher costs, high-quality firms bid similarly to low-quality firms. I use the model to evaluate Rwanda's screening rules. Conditioning promotion on completed work lowers both spending and delays relative to random promotion. A promotion rule aimed at minimizing delays reduces spending by 15.7 percent and delays by 2.6 percent. These results show how conditioning future eligibility on past performance can mitigate the price–quality tradeoff.

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Keywords: public procurement, dynamic auctions, adverse selection, public infrastructure.

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1 Introduction

Public infrastructure (e.g., roads, sewage systems, electricity grids) is critical for economic development (e.g., Dinkelman, 2011; Donaldson, 2018), yet governments frequently struggle to ensure its timely completion: 60% of World Bank infrastructure projects are delayed by one year or more (Espinoza and Presbitero, 2022). Infrastructure projects are awarded by governments through public procurement auctions and carried out by private construction firms with heterogeneous *costs* and *quality*. Governments often cannot identify high-quality firms when awarding contracts, nor enforce timely project execution once contracts are signed. In practice, therefore, quality is non-contractible.

Quality non-contractibility creates challenges for governments seeking to deliver infrastructure at low costs with timely completion. Specifically, if supplier quality is unobserved, the government may face an adverse selection ("lemons") problem (Akerlof, 1970). If low-cost suppliers also provide lower quality, increasing competition by adding bidders worsens expected project quality (Manelli and Vincent, 1995). In a static procurement model with adverse selection, the optimal mechanism sets a floor price below which firms may not bid, which, relative to a simple first-price auction, increases costs but also increases the expected quality of the winning firm (Lopomo et al., 2023). Thus, procurement design faces a tradeoff between price and quality: the instruments that raise the expected quality of the winning firm do so by softening or restricting competition. In public infrastructure markets, however, firms repeatedly compete for public contracts. The prospect of future business can therefore affect firm behavior today. The literature on relational contracting shows how buyers can use the value of future opportunities to improve quality in settings with weak contract enforcement. Preserving these incentives may require favoring established suppliers or restricting competition (Board, 2011; Calzolari and Spagnolo, 2026).

A common policy used by governments to leverage the prospect of future business is to condition access to public contracts on suppliers' past performance.[1] In particular, to compete for larger contracts, firms often need to demonstrate proof of completion of smaller projects. These rules reduce the number of firms eligible to compete for larger contracts, possibly making them more profitable. Firms that complete projects faster can build a track record and qualify for these larger contracts sooner. This may allow the government to screen suppliers indirectly, even when quality is unobserved at the award stage. They also affect bidding: winning the auction provides an opportunity to build experience required for future access, which can make firms willing to accept lower payments today. If this opportunity is more valuable to faster firms, they may bid more aggressively despite higher construction costs. These incentives may therefore shift contracts toward higher-quality firms without raising procurement prices.

In this paper, I study whether conditioning future eligibility on past performance can mitigate the

[1]The World Bank's Standard Procurement Document for Works makes experience a pass/fail qualification criterion, requiring a minimum number of similar contracts "satisfactorily and substantially completed" (Section III, criterion 4.2(a)), and EU Directive 2014/24/EU permits requiring "a sufficient level of experience demonstrated by suitable references from contracts performed in the past" as a requirement for participation (Art. 58(4)). Registry systems closer to Rwanda's – eligibility criteria that restrict the size of projects a firm may bid for – exist in countries such as Kenya, Tanzania, Zambia, South Africa, Singapore, Spain, and the Philippines.

price–quality tradeoff in Rwanda's public procurement of infrastructure. I quantify how eligibility rules affect supplier selection, bidding, and competition for public contracts. Public procurement is a large share of Rwanda's economy, accounting for approximately 11% of GDP in 2017/18 (World Bank, 2020). Although the infrastructure sector makes up only 13% of awarded contracts, it accounts for more than half of procurement value annually (Rwanda Public Procurement Authority, 2023). Construction firms repeatedly compete in this market – 83% of contract value goes to suppliers that have already executed an earlier contract. The typical firm submits 5 bids, wins 1 contract, and remains active for 9 quarters. Beyond these repeated interactions, the government accounts for a large share of firms' business: the median contractor earns 66% of its sales from public institutions. In this market, the government conditions access to larger contracts on past performance through a two-sided categorization system. Projects and construction firms are assigned to a category on an ordered ladder that determines which tenders each firm is eligible to bid on. Promotion to a higher category is granted on the basis of past executed contracts. Tenders in the top three categories carry 88% of estimated value but are open to only 12% of firms.

Studying this policy empirically requires observing three objects that are rarely available together: the rules linking past execution to future eligibility, firms' positions and progression along the eligibility ladder, and their performance after receiving a contract. I rely on two novel administrative datasets: procurement records and Value-Added Tax (VAT) declarations. The procurement records are held by the Rwanda Public Procurement Authority (RPPA) in its electronic procurement platform and cover the near-universe of public infrastructure projects between June 2016 and October 2024, including all submitted bids, bidder identities, award outcomes, and signed contracts. I link this data to firms' category promotion histories, which, together with published regulations, allow me to reconstruct each firm's eligibility and trace its progression along the ladder. The procurement records do not track contract completion. I therefore measure execution from tax records, which provide a quarterly panel of government payments for 98 percent of bidders, from 2013 through 2024. Because payments follow certification of completed work, I use the share of each firm's outstanding contract value paid each quarter as a proxy for portfolio completion.

Using these data sources, I first document three empirical facts about this setting that highlight the relevance of the entry eligibility policy and the tradeoff it imposes. First, contract execution delays are pervasive. I document that only 20 percent of firms' project portfolios are fully paid by their scheduled end, and a third remain incompletely paid two years later. Second, portfolio completion speed is a persistent firm characteristic. A firm's completion speed in its first three quarters with outstanding work predicts its subsequent payments: a one-standard-deviation higher predetermined completion score predicts a 10.4 percentage-point higher quarterly completion share in later quarters. I also show that most of the variation in project outcomes is a product of the firm, rather than of the project or the buyer: once portfolio size is held fixed, buyer and match characteristics account for little of that variation in completion speed, while firm fixed effects raise the explained share from 19 to 43 percent. Third, although faster completion is not used to award contracts directly, the category ladder converts completed experience into access to larger

projects. Contracts are awarded via a first-price sealed-bid auction among eligible bidders. Therefore, conditional on the bid rank, firm's past experience and completion speed do not predict contract award. But firms' completed contracts and completion speed predict their likelihood of category promotion. I show the positive and increasing relationship between firms' completed contract value, measured by payments data, and their likelihood of upgrading. Promotion in turn expands access: the median tender in the top category is more than one hundred times larger than in the bottom category, and, within the same firm, sector, and fiscal year, tenders pursued after promotion are 73 percent larger.

These facts show that completion performance is associated with greater future access, but do not establish how much the ladder improves supplier selection or how it changes procurement costs for the government. Promotion depends on past completion, which in turn depends on both the speed of project completion and the amount of outstanding work for other awarded contracts. Bids, which combine construction costs, markups, and the value of future access, may also depend on outstanding work, as well as on firm type. Assessing the consequences of eligibility restrictions therefore requires disentangling these forces and predicting how bidding and contract allocation would change under alternative rules. I therefore specify and estimate a dynamic auction model of this market. The model builds on the dynamic procurement auctions of Jofre-Bonet and Pesendorfer (2003) and De Silva et al. (2023), in which today's award changes the state from which a firm bids tomorrow. In the model, construction firms differ in a permanent, privately known quality type, carry a backlog of unfinished work, and bid for tenders whose eligibility depends on their category. A firm's type governs the rate at which it completes awarded work, and therefore the speed at which the value it wins becomes the paid experience. Each firm carries an individual state: the category that determines which tenders it may enter, the unfinished work (backlog), and the paid experience it has accumulated (which matters for promotion). In each quarter a firm observes the tenders for which it is eligible and may participate in one or more. For each tender it participates in, the firm submits a bid knowing its own cost draw and its rivals' public states, but not their types. Winning therefore imposes a tradeoff: it delivers current revenue and, in the future, may move the firm up the category ladder, but it also adds unfinished work that may raise its costs and lower its participation in future auctions until that work is completed. The firm's optimal bid balances the two, and takes the form of a construction cost plus a static markup, less the discounted gain in continuation value.

Estimation proceeds sequentially, following the two-step approach of dynamic structural models (Hotz and Miller, 1993; Bajari et al., 2007; Aguirregabiria and Mira, 2007). The first step recovers the policy environment and firms' types. I estimate the RPPA's category promotion policy, based on the empirical relationship between accrued paid experience and category promotion, and I treat it as an exogenous rule that firms account for when forecasting how winning an auction changes their eligibility for future tenders. Because not every firm bids in every auction for which it is eligible, I also estimate a descriptive firm participation policy that matches firms to auctions based on firm- and project-level observable characteristics. I assume two firm types, *High* and *Low*, and assume

that high-type firms complete and are paid for their outstanding work at a faster rate.

The challenge in recovering firm types is that quality and the state are likely correlated. A firm's type affects the states it reaches: faster completion reduces backlog and builds the paid experience needed to move up the category ladder. Its observed completion shares, participation, and bids therefore reflect both its type and its state. I use the panel structure of my data to separate the two, as I observe the same firm over many quarters, tracking its state and three key margins in every quarter – its share of outstanding work completed, its participation in auctions, and the bids it submits. I assume that, conditional on the firm's type and its state, transitory shocks to these three margins are independent mutually and across quarters; that is, persistence in these three margins beyond what is explained by the firm's state is attributable to its type (Hall and Zhou, 2003; Kasahara and Shimotsu, 2009). This strategy allows me to capture permanent unobserved differences in firm quality, but it rules out any effort choice by firms that affects project completion, as in moral hazard models of procurement (Laffont and Tirole, 1987; Lewis and Bajari, 2014), as well as learning-by-doing in project execution. I estimate the firm type mixture and the type-specific policies jointly using the Expectation–Maximization algorithm of Arcidiacono and Miller (2011), which alternates between assigning each firm a probability of being high-type given its history, and re-estimating the completion process and the participation and bid policies of each type with those probabilities as weights, until both converge.

The second stage recovers the primitives that govern firms' bidding behavior. At the auction stage, firms observe rivals' identities, but not rivals' types. They form beliefs about rivals conditional on public states by updating a prior of the (category-specific) rival types using the information from observing the realized set of entrants. [2] Using these beliefs, I construct hazard functions for each bidder in each auction, which summarize the level of competition in that auction and determine the static markup. Given the beliefs and the estimated policies from the first stage, I compute the continuation value of winning from firms' conditional choice probabilities (Hotz and Miller, 1993; De Silva et al., 2023). Each observed bid then decomposes into three parts through the first-order condition (Guerre et al., 2000; Jofre-Bonet and Pesendorfer, 2003): the static markup, pinned down by how steeply a firm's probability of winning an auction falls with its bid; the continuation gain, pinned down by the gain to the firm's future of winning the auction, incorporating the effects of backlog, paid experience, and the probability of category promotion; and the construction cost, which is the residual after computing the first two components. The type-specific cost parameters follow from the distribution of costs across bids. This estimation stage yields structural cost primitives, static markups, and dynamic markdowns by firm type.

I estimate that around 20% of firms are of the high-quality type. At the median backlog, high-type firms complete 42% of their outstanding project work per quarter, compared with 22% for low-type firms. High-type firms convert awards into completed work and gain access to larger contracts faster than low-type firms. In the estimated model, winning is therefore more valuable to

[2] This restriction on beliefs keeps the model tractable but rules out learning of others' types over time, and is similar to that in Backus and Lewis (2025).

high-type firms, inducing them to bid more aggressively despite their higher costs. Comparing firms in the same state bidding for the same project, high-type firms' estimated construction costs exceed those of low-type firms by 43.7% relative to the project cost. However, their larger continuation gain reduces bids by an additional 37.7% of that estimate. Thus, the dynamic markdown offsets about 86% of their estimated cost disadvantage.

I use the model to evaluate two sets of counterfactual policies. First, I relax category eligibility restrictions. Modestly relaxing the requirements – allowing firms one category below to the smaller half of projects one category above theirs – reduces spending on commonly awarded lots by 2.2%. Opening all projects to firms one category below reduces total procurement spending by 6.2% and increases value-weighted delays by 4.7%.

Second, I compare the status quo with alternative promotion rules to understand how the RPPA's promotion rule based on past performance compares with alternative policies. As a benchmark, I consider a naïve rule in which the government promotes firms at random within each category. This rule breaks the link between winning a tender today and gaining greater market access in the future. The RPPA's existing promotion rule improves outcomes relative to random promotion, reducing spending by 7.7% and value-weighted delays by 2.3%. However, I find that the government could have achieved even better outcomes with a rule that promoted more firms. I find that a rule aimed at minimizing delays doubles total firm promotions, and triples promotions into the highest category, expanding access to the largest contracts. This rule lowers value-weighted delay by 2.6% and spending by 15.7% relative to the status quo. The gains from screening therefore depend on how the promotion rule rewards performance and affects competition.

This paper contributes to the literature on procurement in weak contracting environments, where quality is not fully contractible. Theory highlights how adverse selection can make price competition favor low-quality suppliers, and how procurement design can trade off lower prices against higher expected quality (Manelli and Vincent, 1995; Lopomo et al., 2023). The empirical literature has focused on contract design, competition, and procurement rules (Lewis and Bajari, 2011; Bajari et al., 2014; Decarolis, 2014; Carril et al., 2026), as well as imperfect enforcement and buyer discretion (Coviello et al., 2018; Ryan, 2020, 2026; Decarolis et al., 2025). The closest paper is Andreyanov et al. (2024), who study scoring auctions that reward audited past performance and find that the policy improves contract performance without increasing prices. Like the rest of the empirical literature on this tradeoff, their analysis uses a static auction model. In contrast, I study the price–quality tradeoff in a dynamic setting, where completing work today affects access to future contracts. I use a dynamic model to quantify how the value of these future opportunities changes current bidding and supplier selection, and hence procurement prices and project completion.

Methodologically, I build on the dynamic auction model of Jofre-Bonet and Pesendorfer (2003). I follow De Silva et al. (2023) in formulating the model and recovering continuation values in a moment-based equilibrium, given the large number of firms in this market. I extend this framework to incorporate persistent unobserved contractor types that differ in construction cost and project completion speed, as well as an eligibility ladder that links completed work to future opportunities.

I use the expectation–maximization algorithm of Arcidiacono and Miller (2011) to estimate latent firm types jointly with bidding, participation, and completion behavior.

The paper also relates to the literature on relational contracting under limited contract enforcement. Theoretical work has shown that the value of future business can make it optimal for buyers to favor established suppliers or restrict competition (Board, 2011; Calzolari and Spagnolo, 2026). Evidence from low-income countries shows how future business sustains supplier relationships and how competition affects their value (Macchiavello and Morjaria, 2015, 2021). Within this literature, the closest work is Narayanan (2025), which studies a private buyer that promotes suppliers into tiers with greater order volumes in a model that incorporates adverse selection and moral hazard. Relative to this literature, I focus on the government as a buyer and study formal government eligibility rules that determine access to future contracts, while first-price auctions determine awards and prices. My model isolates selection across persistent quality types and bidding responses to future market access, abstracting from an effort choice.

Finally, the paper relates to the broader literature on state effectiveness and public goods and service delivery in low- and middle-income countries. Research in this literature shows how public-sector performance depends on bureaucratic management and capacity (Rasul and Rogger, 2018; Bandiera et al., 2021; Best et al., 2023), oversight (Olken, 2007), and political incentives and contractor selection (Lehne et al., 2018; Rogger and Somani, 2023; Brugués et al., 2024). Work on infrastructure specifically examines how procurement procedures and contracting arrangements affect quality and timeliness (Lewis-Faupel et al., 2016; Wolfram et al., 2026). I complement this work by examining how governments' eligibility rules select among contractors with persistently different completion speeds.

Section 2 describes Rwanda's public works procurement market, the categorization policy, and the data, including the construction of the payment-based measure of contract execution from VAT withholdings. Section 3 documents the three empirical facts. Section 4 develops the dynamic auction model, Section 5 describes estimation, and Section 6 presents identification. Section 7 reports the structural estimates, and Section 8 the counterfactuals: relaxations of eligibility followed by changes to the promotion rule. Section 9 concludes.

2 Setting and Data

2.1 Public Works Procurement in Rwanda

Public procurement is the primary instrument of infrastructure provision in Rwanda. Public Works tenders[3] are administered by procuring entities (PEs)—ministries, districts, hospitals, schools, and other public bodies—each of which advertises and administers its own tenders under the regulation of the Rwanda Public Procurement Authority (RPPA). The suppliers are construction firms, the

[3]A *tender* is an auctioned project (e.g., the paving of a highway segment). A tender may be subdivided into *lots*, which are advertised together but bid on and awarded separately; the lot is therefore the unit of competition, and I treat each lot as an independent auction throughout. Because 85 percent of tenders contain a single lot, the text refers to auctioned projects as tenders rather than lots when the distinction is immaterial.

overwhelming majority of which (more than 95% in the bidding data described below) are registered as formal firms in Rwanda.[4] Throughout the paper I focus on the three main Works sectors – Buildings, Roads and Bridges, and Drinking Water Supply –, which constitute 90% of all tenders in my sample.

A Works tender proceeds through a sequence of steps recorded in my data. The PE advertises the tender—divided, where appropriate, into lots—on *Umucyo*, Rwanda's electronic procurement platform, which the RPPA launched nationally in 2017. Umucyo covers roughly 85–90 percent of public tenders; the remainder are tendered offline and are not observed in my data. Firms submit sealed bids, alone or as part of a joint venture (JV) consisting of several firms. Bid evaluation proceeds in two stages: first, bids that do not meet administrative and technical responsiveness criteria are disqualified. Second, among the qualified bids, the contract is then awarded to the lowest evaluated price. After award, the winning firm signs a contract specifying a final contract value and a start and end date. Subsequent changes to value or duration are recorded as contract amendments. During execution, the firm files an invoice upon completing each portion of the works, the PE certifies the invoice, and payment is released. At the moment of payment, public institutions withhold Value-Added Tax (VAT). This institutional feature underlies my measurement of contract execution, and I return to it below. The RPPA additionally maintains a public blacklist of firms which commit serious breaches of contract and are fined or barred from future tenders.

Three features make this setting well suited to studying quality screening. First, electronic procurement records track the near-universe of Works tenders from advertisement through amendment. Second, tax records can be used to measure when firms are actually paid for Works contracts. Third, experience-based eligibility rules that govern market access are public, and can be estimated using the first and second components. I now turn to these rules, which create the supplier categorization system.

Supplier Categorization System Since 2014, public procurement of infrastructure (Public Works) in Rwanda uses a formal category system to determine firm eligibility for each tender. The RPPA's categorization manual motivates the system by noting that, without it, firms could win "tenders of higher value compared to their capacity" (RPPA, 2014, p. 5). It also describes the category structure as facilitating firms growth into higher categories (p. 8). In conversations, high-level RPPA officials confirmed these two objectives for the categorization policies. First, and most importantly, to reduce delays and project abandonment of large and complex projects by firms without the capacity to complete them. Second, to reserve smaller projects for smaller firms to gain experience and grow. The advent of electronic procurement in 2016–17 in Rwanda facilitated the widespread adoption and accurate application of the categorization system by every PE, as well as stringent enforcement of firm eligibility criteria. The system is generally seen as a success within the RPPA, although questions remain about the degree to which it encourages firm growth due to

[4]This share includes subsidiaries of multinational firms owned (or partially owned) by foreign firms. Unfortunately, my data does not include ownership information, so I am unable to accurately distinguish between foreign and Rwandan firms.

low rates of firm promotion.

The categorization system classifies tenders and firms. When advertised by the commissioning PE, each tender is assigned to an ordered category ($A > B > C > D > E > F$) based on the project's engineering cost estimate (ECE). Each category corresponds to a range of project ECE values. Category F contains the smallest projects and category A the largest. For example, category F covers projects below RWF 100 million, category D covers projects between RWF 300 million and RWF 800 million, and category A covers projects above RWF 2 billion (Rwanda Public Procurement Authority, 2014, pp. 9–12). Given an ECE, the classification is automatic through the electronic system, unless in rare occasions where the PE official requests an alternative categorization.[5] In these cases, the PE must submit justification their alternative categorization and the RPPA must approve it. On the other side of the market, firms are also classified in categories (also A to F). When registering on the Umucyo system to compete for public procurement tenders in a given sector (e.g., Buildings), a firm can request to be categorized in A, B, C, D, or E, or will be placed in category F by default. The RPPA's decision to approve firms' categorization requests is based on four criteria for which the firm must submit documentation: 5-year previous revenue, number and value of previously executed public contracts, owned equipment, and permanent professional staff hired at the firm. Moving up the category ladder requires meeting higher thresholds on every margin. The requirements for a given category are broadly similar across sectors. Appendix Table 6 reports the turnover and past executed-tender requirements for the three sectors studied in the paper. During the sample period, categories D and E required one executed tender, while categories A–C required two, each worth at least 60 percent of the prospective tender value (Rwanda Public Procurement Authority, 2014). [6]

The category system determines bidding eligibility. During the sample period studied in this paper (2016–2023), a firm could bid for tenders in its own category and up to two categories below. For example, a category B firm could bid for projects in categories B, C, and D. The eligible set for a lot is determined by the lot category and each firm's registered category at the time of bidding.[7] Eligibility is sector-specific, and firms are categorized separately for each sector they register in. A firm may hold one category in Buildings and a different category in Roads and Bridges. Figure 1 shows how firms' categories change between their first and last observations in the sample. Among the 394 firms represented in the figure, 18.3 percent are promoted in at least one of their sectors in the sample. The probability of upgrading is higher among firms initially in the top half of the ladder (categories A–C).

[5] For instance, if the project is more complex than the ECE suggests. Miscategorization in either direction occurs for 3.9% of projects in our sample.

[6] In Roads and Bridges, categories A and B are subdivided into $A1/A2$ and $B1/B2$, respectively; I collapse these into A and B for simplicity.

[7] In October 2023 the cross-category rule was tightened. Firms could continue to bid for tenders in their own category, but to bid in lower-category tenders, they must joint venture with a firm in the tender's category.

Figure 1: Category progression in the sample



Notes: The unit is a firm-sector pair in the sample. Rows show the first observed category and columns show the last observed category through October 2023. Cells report row shares; row labels report the number of firm-sector pairs. The sample contains 465 firm-sector pairs from 394 bidders. Categories A and B combine the Roads and Bridges subcategories A1/A2 and B1/B2, respectively.

2.2 Data

I use administrative records from two institutions. From the RPPA, I use the records of its electronic procurement platform, *Umucyo*, for the period September 2016 – October 2024: the procurement records, covering advertisement, bid submission, evaluation, award, contract signing, and contract amendments, and the categorization request records. From the Rwanda Revenue Authority (RRA), I use VAT declaration data covering 2013–2024, from which I recover a firm-quarter measure of government payments, as described below.

The two sources are linked by firms' Taxpayer Identification Numbers (TIN). The procurement records define the auctions, eligible firms, signed commitments, and contract schedules; the VAT records define the realized payment timeline. I supplement these with Rwanda's Open Contracting Data Standard (OCDS) records for 2016–2024, which identify public contracts, supplier IDs, contract dates, and contract values after the tendering process. I use the OCDS records to identify the non-Works (Goods, Services, and Consulting) contracts held by the Works firms in my sample and to net the associated payment flow out of the VAT measure of Works execution.

2.3 RPPA Procurement Records

Between June 2016 and October 2024, Umucyo records 2,211 Works tenders (2,493 tender-lots). The RPPA Umucyo data represents the near-universe of procured Public Works tenders during the sample period. For each tender, Umucyo records the tender notice, tender lots, submitted bids, bidder identities, technical evaluation records, selected winners, signed contracts, and amendments.

The three sectors named above collectively make up 90% of projects: Buildings (60%), Roads and Bridges (16%), and Drinking Water Supply (12%). I restrict the sample to projects awarded through low-price sealed-bid auctions (National or International Competitive Bidding), consisting of 85% of lots. After these sample restrictions, I retain valid bids submitted by lead bidders and joint-venture partners. A contract is signed after about 90 percent of auctions.[8]

From the raw procurement records I construct a tender-lot-bid panel. For each tender-lot, the panel links project characteristics (sector, engineering estimate cost), the procuring entity, bidder (anonymized) TINs, bid prices, joint venture members, evaluation outcomes (disqualifications), selected winner, signed contracts, contract value, contract start and end dates, and contract amendments. To this panel, I link the bidder's registered category at the time of bid submission, as described below.

Firm Category Histories I construct firm-sector category histories from firms' category requests submitted to the RPPA. The records identify the firm TIN, request and response dates, sector, requested category, and approval decision. I convert approved requests into a monthly running category for each firm-sector, which I use to assign bidder eligibility at the bid date and to measure category upgrades. Appendix Table 7 shows that category eligibility requirements are followed during my sample period: 99.5 percent of bids come from firms in the tender category or no more than two categories above it.

2.4 RRA VAT Declarations and Government Payments

While the RPPA data contains detailed information from tender advertisement to tender award, it does not contain detailed information about contract execution. Because payment follows the certified-invoice process described in Section 2.1, government payments trace realized execution. To measure the speed of contract execution, I therefore use data on government payments to firms from Value-Added Tax (VAT) declarations from 2013–2024. Public institutions retain 18 percent VAT on taxable supplies under public tenders, and the retention is automatically recorded when firms file VAT declarations. I measure government payments to firms from the VAT retained by public institutions (Rwanda Revenue Authority, 2025), and construct VAT-implied government payments for firm i in quarter t as

$$Y_{it}^{\text{tot}} = \frac{\text{VAT retained by public institutions}_{it}}{0.18}.$$

Firms file VAT monthly or quarterly, depending on firm size; I aggregate monthly filings to the quarterly payments. I am able to match 98% of unique bidders in the procurement dataset to the VAT data. This yields a firm-quarter panel of government payments from all sources.

[8] Some tenders are closed without a winner being selected. In a small number of cases, a winner is selected but no contract is signed.

2.5 From Payments to Execution and Experience

Allocating payments to the Works portfolio. The VAT payment data record the total payments across all contracts in each time period; they do not identify the contract associated with each payment. It is common for firms to have multiple ongoing contracts: in more than half of firm-quarter pairs in which a firm has an ongoing Works contract, the firm also has another Works or non-Works contract. I therefore combine the VAT data with the timing and value of observed Works contracts from Umucyo to construct a firm-level measure of payments attributable to the Works portfolio.

I describe this process in detail in Appendix A.2. In summary, I construct the total amount of government payments by firm-quarter from all sources and subtract from it the payment amount that cannot be attributed to Works contracts. To measure this share unattributable to Works, I use the value of non-Works contracts from OCDS records, and impute the value of offline tenders that I do not observe in my data.[9] A firm's open *portfolio* in a quarter is the value of Works contracts it has not yet been paid for, i.e., its *backlog* carried over from the previous quarter plus the value of new Works contracts awarded to it that quarter. Payment for Works contracts is defined as the full remaining amount, up to the value of the firm's open portfolio. In each period, the value of Works payments reduces the portfolio; whatever remains unpaid carries over as the next quarter's backlog.

Among the 4,674 firm-quarters with open portfolios, Works payments are positive in 44 percent. Conditional on a positive payment, the median is RWF 32 million and the mean is RWF 225 million.

Portfolio execution and experience. For firm-quarters with an open portfolio, I measure the quarterly portfolio *completion share* as Works payments received in the quarter divided by the value of the open portfolio. I use this payment-based share as a proxy for the proportion of outstanding work completed during the quarter. It ranges from zero, when none of the portfolio is paid that quarter, to one, when all of it is. As firms complete work and receive payments, they accumulate *paid experience*, the cumulative value of Works payments they have received. Firms' promotion up the category ladder depends on the total value of paid experience. Across firm-quarters with an open portfolio, paid experience has a median of RWF 37 million and a mean of RWF 753 million; over the panel, firms accumulate RWF 462 billion of it.

2.6 Validation

I validate the measure of Y^W as payment completion using independently collected project reports submitted by third party firms hired to supervise the contract execution and approved by the procuring entity. A field team collected and coded available project reports and handover documents for 196 tenders. Among these, 124 contracts contained a dated final report in the relevant sectors: 67 are Buildings, 22 are Roads and Bridges, and 35 are Drinking Water Supply. While this sample is

[9]The value of unaccounted payments changes measured Works payments in only 15.6% of quarters when firms have active Works contracts. Moreover, it affects relatively few firms: setting unaccounted payments to zero would increase attributed Works payments by RWF 106.5 billion, but five firms account for 81% of this difference.

selected based on report availability – projects are larger and recent – they provide an independent measure of project completion and delays that is useful to validate the main measure using VAT payments data.

Completion and payment timing. VAT payments are positively associated with project completion. Among 108 contracts for which both reported completion and contemporaneous payments are observed, a 10 percentage point increase in reported physical completion is associated with a statistically significant 5.1 percentage point increase in cumulative Y^W, conditional on contract value, scheduled duration, sector fixed effects, and report-year fixed effects (Appendix Table 5).

For a small sample of 11 contracts, the timing of the firm's other contracts allows us to attribute payments directly to the focal project. The share paid at least 75% of contract value rises from 18% in the recorded final-report quarter to 73% one quarter later and 82% two quarters later. Although the sample is small and potentially selected, payments rise sharply in the quarter after the recorded final report (Appendix Figure 11).

Delays. I measure delays by comparing the report date with the contract's scheduled end date. Among 121 contracts for which we collect report dates, delayed contracts are 13.3 percentage points more likely not to have reached 90 percent payment one quarter after the scheduled end date (standard error 9.5 percentage points). This coefficient is positive, as expected, but not statistically significant (Appendix Table 5). Taken together, the results support interpreting Y^W as a payment measure associated with physical progress, with payments following documented completion with a lag.

3 Empirical Evidence

This section presents three facts about project execution and market access in Rwanda's infrastructure public procurement market. First, payments on public Works contracts frequently lag behind their scheduled timeline, and some contracts remain unpaid well after their scheduled end. Second, portfolio completion speed differs persistently across firms: performance in a firm's first three observed quarters predicts its subsequent payments. Third, the category ladder converts experience, i.e., completed projects, into access to larger tenders. Paid experience predicts promotion, higher categories contain larger projects, and firms win larger contracts after promotion. Taken together, these facts motivate a model with persistent firm types, backlog, accumulated paid experience, and category-dependent market access.

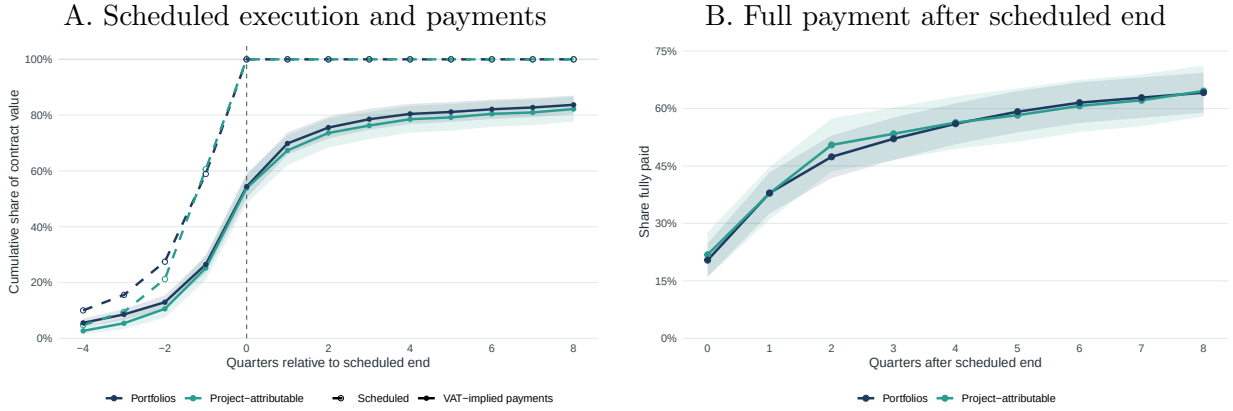
3.1 Fact 1: contract execution delays are pervasive

To measure delays, I use data on the realized contract payment amount and timing (from the VAT data) and the scheduled payment end date (from the procurement contract data). In Panel A of

Figure 2, I plot the scheduled payment timeline assuming payments were equally spaced over the contract duration, and compare it to the realized payment timeline. I do this for both the portfolio and the project-attributable samples, which yield nearly identical results. The graph illustrates that, by the scheduled contract end date, firms have received approximately 54 percent of contract value, 74 percent two quarters later, and 82 percent after eight quarters.

However, the average payment timeline hides significant delays and project non-completion. Panel B reports the share of contracts or portfolios that have reached full payment. At the scheduled end, only 22 percent of project-attributable contracts, and 20 percent of portfolios, are fully paid. The shares rise to 50 and 47 percent after two quarters, and to 65 and 64 percent after eight quarters. Almost one-third of portfolios are not fully paid, highlighting that delays are not driven only by small unpaid balances on otherwise completed portfolios. These facts suggest that public procurement contracts in Rwanda suffer from substantial delays and non-completion.

Figure 2: Payments around the scheduled end



Notes: Panel A reports cumulative scheduled execution and VAT-implied payments relative to the scheduled end. Panel B reports the share of contracts or portfolio episodes that reach full payment. The sample contains 206 project-attributable contracts and 382 portfolio episodes. Shaded bands are 95 percent confidence intervals clustered by firm.

3.2 Fact 2: portfolio completion speed is a persistent firm characteristic

I next ask whether portfolio completion speed contains a persistent firm component by testing whether a firm's completion shares in its first few periods predict its later outcomes. For each firm, I construct a completion score from its first three quarters with outstanding work (i.e., positive backlog). I first residualize the quarterly completion share on the value of work available for payment, new awards, the number of active contracts, and calendar quarter fixed effects. I then average these residuals within firm, shrink the average toward the sample mean, and standardize it. Panel A of Figure 3 shows that the score has wide dispersion across the firms that win at least one contract.

Panel B shows that the (predetermined) completion score predicts several future completion outcomes relatively far into the future. A one-standard-deviation higher predetermined score predicts

a 10.4 percentage-point higher quarterly completion share in later quarters. The estimate remains 8.7 percentage points at least four quarters after the score is fixed, and the probability of a positive completion share is 7.8 points higher. In the project-attributable sample, the same score predicts a 7.4-point higher project payment share and an 11.7-point higher probability of full payment by eight quarters after the scheduled end. The persistence across time and across subsequent contracts is suggestive of heterogeneity in firm-level execution quality.

I provide further evidence that execution speed is largely a firm-specific attribute by testing where variation in quarterly portfolio completion shares comes from. In Appendix Table 9, I report results of regressions of quarterly portfolio completion shares, where I progressively add controls to assess the change in adjusted R^2. Calendar quarter fixed effects alone produce an adjusted R^2 of 0.034. Adding the size and composition of the firm's active portfolio raises it to 0.145. Procuring entity composition and the procuring entities' prior completion performance only raise the adjusted R^2 only to 0.149. Observable measures of firm size and experience raise it to 0.193, while firm fixed effects raise it to 0.431. Thus, workload and observable firm characteristics explain some variation in completion shares, but buyer characteristics explain little; the remaining variation contains a substantial firm-specific component.

Figure 3: Predetermined completion score and subsequent completion



Notes: Panel A plots the standardized, empirical-Bayes-shrunk mean residual completion share in each firm's first three positive-backlog quarters. The sample contains 319 firms; the dashed line marks the mean. Panel B reports the change in subsequent outcomes associated with a one-standard-deviation higher score. Bars are 95 percent confidence intervals clustered by firm.

3.3 Fact 3: the category ladder converts completed experience into access

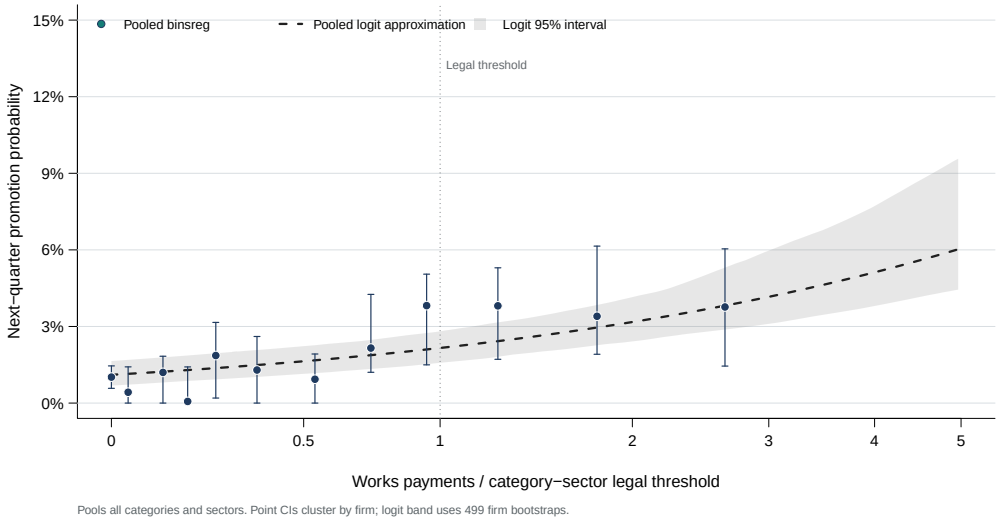
Because contracts are awarded through first-price auctions, execution delays and project non-completion are not used directly in bid evaluation and award. Appendix Table 10 shows that the portfolio completion score does not predict bid disqualification. It also does not predict award conditional on qualification and bid rank.

However, project completion is crucial for determining a firm's future eligibility through the category ladder. As described above, a firm's category determines the size of tenders for which a

firm is eligible.[10] Promotion therefore changes the set of projects available to the firm in the future. To examine how completed experience translates into future eligibility, Figure 4 plots next-quarter promotion against cumulative paid Works experience, P_{it}, measured relative to the category- and sector-specific value requirement. Promotion rises smoothly with paid experience. This pattern is consistent with left-censored pre-Umucyo and offline experience, as well as other statutory promotion requirements.

I provide further evidence of the value of experience for upgrading by comparing observably similar firms that apply for a category promotion in the same quarter. Within request-quarter-by-sector-by-origin-by-target cells, and controlling for firm age, prior Works contracts, and prior category requests, applicants whose measured paid Works experience meets the stated value requirement are 15.1 percentage points more likely to be approved (Appendix Table 8).

Figure 4: Paid experience and category promotion



Notes: Points are quantile-spaced bins controlling for Works sector; bars are 95 percent confidence intervals clustered by firm. The dashed line is a pooled logit fit with a firm-bootstrap confidence interval. The sample contains 5,081 firm-quarter transitions for 357 firms in categories F through B, including 96 promotions. The vertical line marks the stated payment-value requirement.

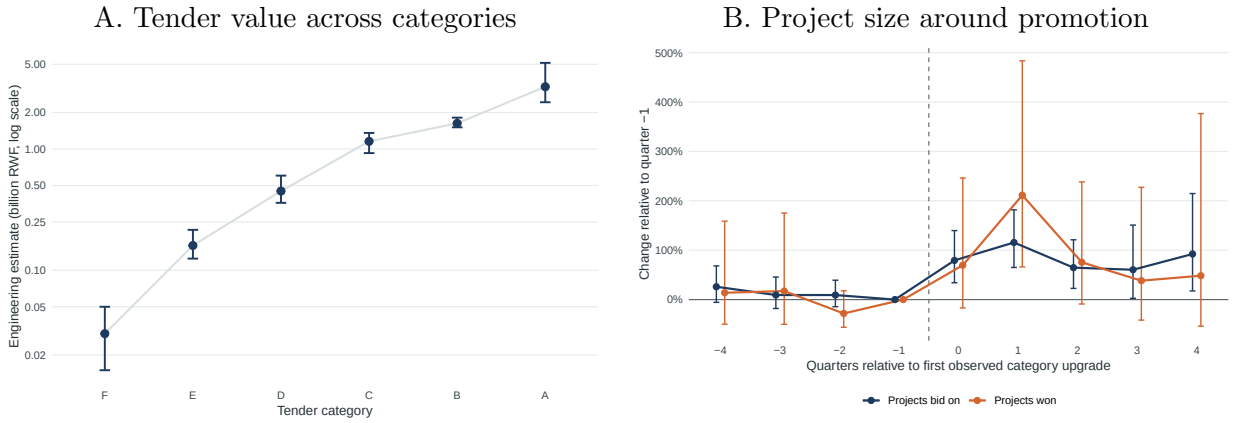
Promotion expands access to larger tenders. Panel A of Figure 5 plots engineering estimates across the ladder. The median tender rises from RWF 30 million in category F to RWF 3.26 billion in category A. The increase is monotone across categories and is two orders of magnitude larger at the top than at the bottom of the ladder.

Firms use the additional access from category promotion. Within the same firm, Works sector, and fiscal year, tenders pursued after an observed category upgrade are 73 percent larger, and winning tenders are 80 percent larger. Panel B plots the corresponding coefficients. Project size is stable over the quarters immediately before promotion and rises when the firm is first observed in the higher category.

[10]The restriction binds in the data: 99.5 percent of classified bids in Buildings, Roads and Bridges, and Drinking Water Supply satisfy the rule that firms bid in their own category or up to two categories below it (Appendix Table 7).

Accordingly, category progression is mostly upward: 20.9 percent of the 465 firm-sector pairs in the sample of regular bidders move to a higher category, 78.3 percent remain in the same category, and only 0.9 percent move down (Figure 1).

Figure 5: Category promotion and access to larger projects



Notes: Panel A reports median engineering estimates and interquartile ranges for 1,710 competitive Works lots before October 2023; values are in billion RWF on a log scale. Panel B reports coefficients relative to quarter −1 from regressions of log engineering estimates on event time, absorbing firm-by-sector-by-fiscal-year fixed effects. Bars are 95 percent confidence intervals clustered by firm.

3.4 Mapping the evidence to the structural model

The three facts describe the RPPA's indirect screening mechanism. Firms differ persistently in their execution speed. While execution speed is not used directly in awarding contracts, it does determine how paid experience accumulates. Paid experience then translates into category promotion, which opens access to larger tenders and potentially increases firm revenue:

$$\text{firm type} \longrightarrow \text{completion of outstanding work} \longrightarrow \text{paid experience}$$
$$\longrightarrow \text{category promotion} \longrightarrow \text{access to larger projects.}$$

Therefore, the category system can select firms with faster project execution into larger markets without requiring the government to observe their type directly.

The empirical evidence motivates three features of the model. Firms carry a stock of backlog, which rises when they win contracts and falls as payments are received. Firms differ in persistent execution speed, represented by type θ_i. As firms complete unfinished work, they accumulate paid experience P_{it}, which affects their probability of upgrading categories and hence the tenders they may participate in.

These facts describe how firms progress through the category ladder, but do not show how the ladder affects bidding. In particular, the descriptive evidence cannot separate construction costs from static markups and the continuation value of future market access. I use the structural model to make this decomposition and to quantify the tradeoff between screening firms through eligibility and increasing procurement prices by restricting competition.

4 Model

4.1 Overview

The model is based on the standard dynamic auction model of Jofre-Bonet and Pesendorfer (2003). Firms participate repeatedly in first-price procurement auctions, winning adds backlog, and bids reflect the continuation value of an award. I extend these models to capture key features of my setting: I add unobserved and persistent differences in firm execution, and a category ladder that conditions future tender eligibility on past paid experience. These features allow me to quantify how the category ladder screens execution types against the loss of competition from restricting eligibility.

In each period, a firm's category determines the tenders for which it is eligible. Firms differ in a permanent private type $\theta_i \in \{H, L\}$, which governs their speed of project completion. As work is completed and paid, firms accumulate paid experience and may move up the category ladder. Execution is stochastic conditional on type and state; firms do not choose execution effort. The RPPA category promotion policy and firms' participation policies are exogenous and assumed to be known by all firms.

Conditional on participation, firms draw private costs and optimally choose their bids. When choosing their bid, firms trade off current-period profits, which depend on their costs and competition in that auction, with the net continuation value of winning, which enters the bid as a dynamic markdown. Winning a contract has opposing dynamic effects: it adds backlog, but completing a project expands the firm's future market access. High-type firms complete projects and accrue experience faster than Low-type firms, so, in general, they have larger dynamic value of winning.

Firms know their own type, but the RPPA and rival firms do not observe it. In the each auction, firms observe the identities of participating rivals and each rival's public state. They form beliefs about rival types by integrating over the type distribution in each category, rivals' public states, and the participation policy. When forecasting future competition, firms do not track the full distribution of rival states. Instead, each firm tracks its own public state and a set of aggregate moments that capture competition in each of the categories, as in De Silva et al. (2023).

4.2 Environment

4.2.1 Firms, Tenders, and Eligibility

Time is discrete and measured in quarters, $t = 0, 1, \ldots$. Public works are divided into sectors $m \in \mathcal{M}$, where $\mathcal{M} = \{\text{Buildings}, \text{Roads}, \text{Water}\}$. Registration categories are

$$\mathcal{K} = \{F, E, D, C, B, A\},$$

ordered from F (lowest) to A (highest); define $R(F) = 1, \ldots, R(A) = 6$, so higher categories have higher ranks. Registration is sector-specific. Firm i's category portfolio is

$$\boldsymbol{\kappa}_{it} = (\kappa_{im,t} : m \in \mathcal{M}),$$

where $\varnothing$ denotes an inactive sector. Observed entry into a new sector is included in the estimated category transition; Appendix C.3 describes its treatment.

Auctions are indexed by ℓ and arrive exogenously in each period. In each quarter, the number of new projects in tender category k is a Poisson draw with intensity λ_k. Each project has public characteristics

$$\chi_\ell = (m_\ell, k_\ell, z_\ell, q_\ell)$$

drawn from a category-specific distribution G_k^χ, i.i.d. across projects and quarters. Here m_ℓ is the sector, k_ℓ the tender's category, z_ℓ the engineering cost estimate (ECE), and q_ℓ other tender characteristics.

For a project ℓ in sector $m = m_\ell$, firm i is eligible if it is registered in sector m and the project lies in its category or no more than two categories below it:

$$Q_{i\ell t} = \mathbf{1}\{\kappa_{im,t} \neq \varnothing,\ 0 \leq R(\kappa_{im,t}) - R(k_\ell) \leq 2\}. \tag{1}$$

Let $\mathcal{I}_t$ denote the active firms in the procurement market in quarter t. These firms have entered, i.e., previously submitted a bid, and have not exited. In each period t, the realized set of firms eligible for project ℓ is denoted $\mathcal{E}_{\ell t} = \{i \in \mathcal{I}_t : Q_{i\ell t} = 1\}$. The function Q_{ilt} implements the RPPA category eligibility requirements, based on the firm and tender category.

4.2.2 Individual and Aggregate States

Firm i has permanent quality type

$$\theta_i \in \{H, L\}.$$

As described in the implementation section below, firm type combines three aspects: auction participation, submitted bids, and portfolio completion speed. I assume H firms have stochastically higher quarterly portfolio completion shares. A firm's type θ_i is permanent and private; neither other firms nor the government observe it.

A firm's individual state is

$$s_{it} = (\boldsymbol{\kappa}_{it}, B_{it}, W_{it}), \tag{2}$$

where B_{it} is the total value of Works projects that remains to be completed, and W_{it} is cumulative awarded Works value. Paid experience, the total value paid to the firm for its completed Works projects, is

$$P_{it} = W_{it} - B_{it}. \tag{3}$$

To forecast future competition, firms use an aggregate state that summarizes recent competition

across auction categories:

$$\bar{s}_t = \left(\bar{n}_t^A, \bar{n}_t^B, \bar{n}_t^C, \bar{n}_t^D, \bar{n}_t^E\right), \tag{4}$$

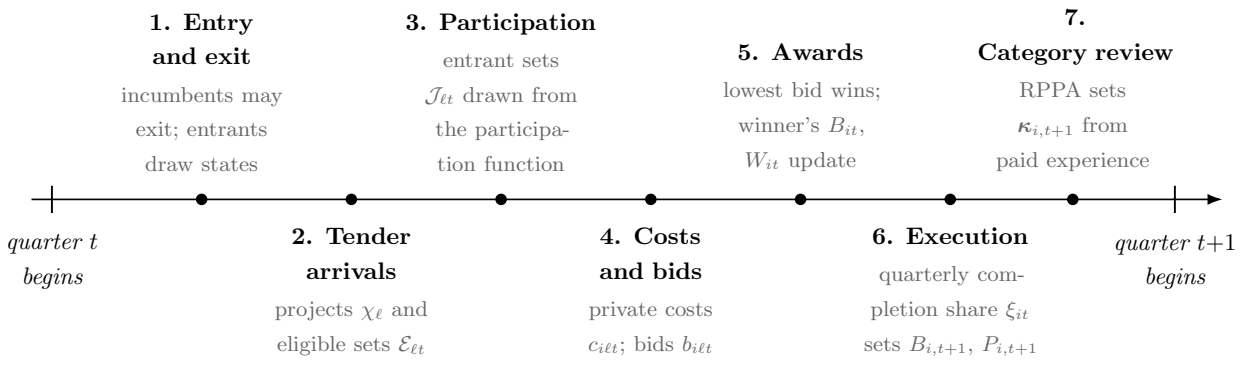
where $\bar{n}_t^k$ is the average number of qualified bidders per category-k auction lot over the preceding four quarters, $t-4, \ldots, t-1$, pooled across the three Works sectors. These measures are known at the beginning of quarter t and capture competition in the categories firms can reach through promotion. When evaluating future opportunities, firms condition on these aggregate measures rather than tracking the full distribution of rivals' individual states, following De Silva et al. (2023). Appendix C.4.3 describes their construction and law of motion.

Each firm therefore tracks $(s_{it}, \bar{s}_t)$.

4.2.3 Industry Entry and Exit

The set of active firms $\mathcal{I}_t$ evolves through exogenous entry and exit. Entrants draw initial states and permanent characteristics from an exogenous distribution that may depend on $\bar{s}_t$. Incumbents face an exogenous exit probability that depends on their individual and aggregate states. Exit is absorbing.

4.3 Within-Quarter Timing



4.4 Execution, Backlog, and Payment Experience

4.4.1 Quarterly Completion Share

Let

$$\Delta W_{it} = \sum_{\ell \in \mathcal{W}_{it}} z_\ell^c \tag{5}$$

be the VAT-exclusive signed-contract value of projects awarded to firm i in quarter t, where $\mathcal{W}_{it}$ is the set of projects awarded to the firm that quarter and z_ℓ^c is contract ℓ's signed value net of VAT. These equations describe the realized states. When evaluating a prospective award at the bidding stage, I use the project's engineering estimate (z_ℓ), rather than the signed contract value, as the increment to backlog and cumulative awarded value. As in existing work Jofre-Bonet and

Pesendorfer (2003); Groeger (2014); De Silva et al. (2023), this assumption makes the increment independent of the firm's bid. The stock of work that can be completed is

$$X_{it} = B_{it} + \Delta W_{it}. \tag{6}$$

For $X_{it} > 0$, the quarterly completion share is

$$\xi_{it} = \frac{Y_{it}^{W}}{X_{it}} \in [0, 1], \tag{7}$$

where Y_{it}^{W} is VAT-exclusive Works payment capped at X_{it}. Quarters with $X_{it} = 0$ are outside the execution risk set.

Conditional on permanent type and post-award stock,

$$\xi_{it} \sim F_{\xi}(\cdot \mid X_{it}, \theta_i). \tag{8}$$

The distribution of quarterly completion shares can depend on X_{it}. All else equal, a firm with more outstanding work will likely complete a lower share of that work in each period. I assume that type H has a stochastically higher completion share:

$$F_{\xi}(x \mid X, H) \leq F_{\xi}(x \mid X, L) \quad \text{for all } x \in [0, 1]. \tag{9}$$

High-type firms complete work and gain experience faster, which increases their probability of category promotion in each period. A firm's type is permanent: conditional on type and outstanding work, accumulated experience does not affect the distribution of completion shares. The model therefore rules out learning by doing in project completion speed; firms cannot improve their completion speed over time. In the empirical implementation, I parametrize the distribution to allow for no completion, partial completion, and full completion in each period.

4.4.2 State Transitions

The states evolve by

$$B_{i,t+1} = (1 - \xi_{it})(B_{it} + \Delta W_{it}), \tag{10}$$

$$W_{i,t+1} = W_{it} + \Delta W_{it}. \tag{11}$$

Paid experience accrues as the firm completes new work:

$$P_{i,t+1} = W_{i,t+1} - B_{i,t+1} = P_{it} + \xi_{it}(B_{it} + \Delta W_{it}) = P_{it} + Y_{it}^{W}. \tag{12}$$

Faster execution reduces next-period backlog and increases paid experience by the same amount. Because promotion depends on paid experience, the transition connects firm type to future eligibility.

The category transition occurs after quarter-t execution:

$$\boldsymbol{\kappa}_{i,t+1} \sim \Pi(\cdot \mid \boldsymbol{\kappa}_{it}, P_{i,t+1}). \tag{13}$$

Thus $\boldsymbol{\kappa}_{it}$ governs eligibility in quarter t. Work completed during the quarter first affects eligibility in quarter $t+1$. As described in Section 5.2, the kernel is estimated from observed category histories and held fixed in the firm's problem. It does not condition directly on θ_i; type affects category progression through paid experience.

4.5 Auction Participation

In each quarter, each firm may be eligible to participate in multiple tenders. At the auction stage, for each eligible firm–project pair, let $a_{i\ell t} = 1$ if firm i submits a bid for project ℓ in quarter t. Among eligible firms, the exogenous participation policy is

$$a_{i\ell t} \sim \text{Bernoulli}\left(a_{\theta_i}(\chi_\ell, s_{it}, \bar{s}_t)\right). \qquad i \in \mathcal{E}_{\ell t} \tag{14}$$

The policy yields, for each project, the realized set of bidders $\mathcal{J}_{\ell t} = \{i \in \mathcal{E}_{\ell t} : a_{i\ell t} = 1\}$. This policy depends on project-level observables, the firm's type, individual state, and the aggregate state. It therefore captures how firms-level observable characteristics and type affect the probability of participation. For instance, it allows for firms to be more likely to participate in tenders in their own category rather than categories below, and for firms with more outstanding work to participate less. I hold the participation function fixed in counterfactuals, but I allow participation probabilities to change with eligibility and firms' simulated states.

4.6 Construction Costs and Bidding

Private Construction Costs Conditional on participation, firm i observes a private project-specific construction cost distributed as

$$c_{i\ell t} \sim F_c(\cdot \mid \chi_\ell, s_{it}, \theta_i). \tag{15}$$

Here $\chi_\ell = (m_\ell, k_\ell, z_\ell, q_\ell)$ and $s_{it} = (\boldsymbol{\kappa}_{it}, B_{it}, W_{it})$, with relevant category $\kappa_{im_\ell,t}$ and paid experience $P_{it} = W_{it} - B_{it}$. Conditional on $(\chi_\ell, s_{it}, \theta_i)$, cost draws are independent across firms, projects, and quarters.

Auction Information Structure and Belief Formation For the remainder of this subsection, set $m = m_\ell$ and let $\omega_{H,\kappa_{jm,t}}$ denote the share of type-H firms in rival j's category κ in sector m at time t. For each eligible rival, firms use the high-type share in that rival's registration category as their prior, then update this belief upon observing participation, as decsribed below.

Assumption 1 (Belief formation)**.** *In each auction ℓ, bidders compete in a first-price sealed-bid auction, where the lowest bid wins. Each firm observes its own $(\theta_i, c_{i\ell t})$. Entrants observe χ_ℓ, the*

entrant set $\mathcal{J}_{\ell t}$, and each rival's public state s_{jt}. Rival types and drawn project costs are unobserved. Conditional on rivals' categories, firm types are independent. Before conditioning on participation, bidders form priors over each rival's type using only that rival's category:

$$\mathbb{P}(\theta_j = H \mid \kappa_{jm,t}, s_{jt}, \chi_\ell, \bar{s}_t) = \omega_{H,\kappa_{jm,t}}. \tag{16}$$

Because the two types participate at different rates, observed participation and rival states provide information about the likelihood. Upon observing participants' identities and states, firms use Bayes' rule to update their prior belief about rival types. The resulting posterior of j's type, given j's participation and individual state, is

$$\tilde{\omega}_{j\ell t} = \frac{\omega_{H,\kappa_{jm,t}} a_H(\chi_\ell, s_{jt}, \bar{s}_t)}{\omega_{H,\kappa_{jm,t}} a_H(\chi_\ell, s_{jt}, \bar{s}_t) + (1 - \omega_{H,\kappa_{jm,t}}) a_L(\chi_\ell, s_{jt}, \bar{s}_t)}. \tag{17}$$

Priors about rival types are determined by the share of high-type eligible firms in the rival categories. Participation is informative because the two types may participate at different rates. If high-type firms are more likely to participate at a given state, observing a firm's participation raises the probability assigned to it being high-type. Public states therefore affect posterior beliefs through participation probabilities, even though the initial prior depends only on category. The assumption rules out learning about rivals' type over time from their public states, while allowing those states to affect behavior (Backus and Lewis, 2025).

Rival Type Distributions. I restrict attention to bidding strategies that are strictly increasing in the firm's (private) cost draw, conditional on its state, type, and the information available at the bidding stage (Guerre et al., 2000; Athey and Haile, 2007). Using the observable information about rival bidders' states and their belief about rival types as in Equation (17), bidder i forms beliefs about bids that each rival j submits. Let $F_{\theta,j}$, $f_{\theta,j}$, and $\bar{F}_{\theta,j} = 1 - F_{\theta,j}$ denote the equilibrium bid distribution, density, and survival function of a type-θ rival j, conditional on χ_ℓ, its observable state s_{jt}, and the aggregate state; I suppress these arguments below.[11]

From i's perspective, rival j's bid has the mixture distribution

$$F_j^{\text{mix}}(b) = \tilde{\omega}_{j\ell t} F_{H,j}(b) + (1 - \tilde{\omega}_{j\ell t}) F_{L,j}(b), \tag{18}$$

and since this is linear in the distribution function, the density and survival function carry the same weights:

$$\begin{aligned} f_j^{\text{mix}}(b) &= \tilde{\omega}_{j\ell t} f_{H,j}(b) + (1 - \tilde{\omega}_{j\ell t}) f_{L,j}(b), \\ \bar{F}_j^{\text{mix}}(b) &= \tilde{\omega}_{j\ell t} \bar{F}_{H,j}(b) + (1 - \tilde{\omega}_{j\ell t}) \bar{F}_{L,j}(b). \end{aligned} \tag{19}$$

This mixture governs i's chance of winning, as described below.

[11] The survival function is the probability that j bids above b; i.e., the probability that i beats j with a bid b.

Single-qualified-bidder Auctions. In the estimation sample, 34.9% of auctions have a single qualified bidder. Similar to a government reserve, I introduce a simulated rival firm that competes with the single qualified bidder on these auctions. This rival is never awarded the contract, but its presence enters the win probability and the hazard as though it could. The simulated rival submits a bid based on a cost that is sampled from the distribution of firm-states in that category. While strong, this assumption allows win probabilities and hazards to remain well-defined for all auctions in the sample.

Winning Probability and i's Hazard. When facing rival bidder set $\mathcal{J}_{\ell t}$, bidder i's probability of winning with bid b is

$$G_i(b \mid \mathcal{J}_{\ell t}) = \prod_{j \in \mathcal{J}_{\ell t} \setminus \{i\}} \bar{F}_j^{\text{mix}}(b). \tag{20}$$

Let firm i's hazard function, the proportional rate at which i's probability of winning falls as i raises its bid, be

$$H_i(b \mid \mathcal{J}_{\ell t}) = -\frac{\partial \log G_i(b \mid \mathcal{J}_{\ell t})}{\partial b} = \sum_{j \in \mathcal{J}_{\ell t} \setminus \{i\}} \frac{f_j^{\text{mix}}(b)}{\bar{F}_j^{\text{mix}}(b)}. \tag{21}$$

This hazard determines firm i's competitive environment in project ℓ: it rises with each additional rival. It captures both the number and types of rivals a firm faces. Holding their bid distributions fixed, more rivals increase the hazard; rivals' states and type probabilities also matter through their bid distributions. A higher hazard means that raising the bid causes a larger proportional fall in the probability of winning.

4.7 Dynamic Problem, Bidding, and Equilibrium

Value Function Let $V_{\theta_i}(s_{it}, \bar{s}_t)$ denote firm i's value at the beginning of quarter t, conditional on remaining in the procurement market and before auction arrivals. Conditional on participation, firms evaluate all bids using the same beginning-of-quarter individual and aggregate states, together with the information observed in each auction. I treat bidding within the quarter as simultaneous. Awards, completion, and category transitions determine the following quarter's states. This timing convention aligns the bidding problem with the quarterly measurement of backlog and paid experience.

Firms can bid in multiple auctions within a quarter: in the estimation sample, 46 percent of the 1,461 bidding firm-quarters involve two or more auction lots. For tractibility, I assume auction profits are additive across auctions in the current period. Hence, I evaluate each contract's continuation gain separately, comparing winning that contract with receiving no new awards that quarter. This assumption implies firms do not account for the effect of winning other auctions in the same period when selecting their bid.

The Bellman equation at the beginning of the quarter is

$$V_{\theta_i}(s_{it}, \bar{s}_t) = \mathbb{E}\left[\sum_{\ell:\, i \in \mathcal{J}_{\ell t}} \underbrace{G_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})}_{\text{prob. of winning}} \underbrace{\left(b_{i\ell t} - c_{i\ell t}\right)}_{\text{profit if win}} + \beta \underbrace{V_{\theta_i}(s_{i,t+1}, \bar{s}_{t+1})}_{\text{next-period value}} \,\middle|\, s_{it}, \bar{s}_t; \theta_i \right] \tag{22}$$

The expectation integrates over the tender arrivals, the firm's exit, auction participation, private costs, rival bids, awards, completion shares, category transitions, exit, and the law of motion for $\bar{s}_t$. If the firm exits before the next quarter's auctions, its continuation value is zero. Exit is permanent and yields zero scrap value.[12] The transition to $s_{i,t+1}$ follows (10)–(13); in particular, an award adds to backlog before execution and expands eligibility after it has been completed and paid (and only in future quarters).

Continuation Value of Winning and Optimal Bids. Consider a project ℓ for which firm i is bidding. Let $s^{\text{win}}_{i,t+1}$ and $s^{\text{lose}}_{i,t+1}$ denote the next-period states when the firm wins and loses project ℓ, respectively, including possible exit before the next quarter's auctions. The two paths differ only in the focal award: winning adds z_ℓ to backlog and cumulative awards before completion. The continuation gain from winning is

$$\begin{aligned} D_{i\ell t} &\equiv D(s_{it}, \chi_\ell, \bar{s}_t; \theta_i) \\ &= \beta \mathbb{E}\left[V_{\theta_i}(s^{\text{win}}_{i,t+1}, \bar{s}_{t+1}) - V_{\theta_i}(s^{\text{lose}}_{i,t+1}, \bar{s}_{t+1}) \mid s_{it}, \chi_\ell, \bar{s}_t; \theta_i \right]. \end{aligned} \tag{23}$$

The expectation in (23) integrates over the completion process, category transitions, the law of motion for $\bar{s}_t$, and next-period exit, using the same transition policies for both paths. Winning an auction brings about the accumulation of paid experience and a higher probability of category promotion. But, during contract execution, it also raises backlog, which lowers auction participation and may change contruction costs.

The continuation gain depends on firm type through completion speed and future profits. Holding the firm's individual state fixed, the stochastic ordering of completion shares (equation (9)) implies that high-type firms leave less backlog and accumulate more paid experience on average. This increases their probability of promotion and can make winning more valuable by bringing forward access to larger projects and reducing the burden of additional backlog. However, $D_{i\ell t}$ compares winning with losing, and faster completion affects both paths. Whether high types have larger continuation gains also depends on their current state and expected profits from future contracts. The ordering of completion shares alone therefore does not imply $(D_H > D_L)$; I quantify this difference using the model.

Conditional on the entrant set and the public states of its rivals, the terms in (22) that depend on firm i's bid in project ℓ reduce to

$$G_i(b \mid \mathcal{J}_{\ell t}) \left[b - c_{i\ell t} + D_{i\ell t} \right], \tag{24}$$

[12] In principle, this normalization is not innocuous, and may bias estimation of other primitives (Aguirregabiria and Suzuki, 2014). Furthermore, some counterfactuals (those that change firms' choice sets or the environment's transition structure) are also not identified without knowledge of the scrap value level (Kalouptsidi et al., 2017, 2021). In practice, the normalization is likely to be second order in this setting. Firms that exit almost always continue filing taxes with positive sales in the private market, meaning exit from the procurement market means redirecting capacity towards private market activity rather than firm death.

where $G_i(b \mid \mathcal{J}_{\ell t})$ is as in Equation (20). The full derivation is presented in Appendix D.2. At an interior optimum, differentiating (24) with respect to the bid and using the hazard in (21) yields

$$b_{i\ell t} = c_{i\ell t} \quad + \quad \underbrace{\frac{1}{H_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})}}_{\text{static markup}} \quad - \quad \underbrace{D_{i\ell t}}_{\text{dynamic markdown}}, \tag{25}$$

where the static markup is the inverse of the hazard in equation (21), and the dynamic markdown is as in equation (23). Conditional on the firm's type and auction information, (25) recovers the cost associated with each observed bid, given the estimated hazard and continuation gain. The static markup reflects competition in the current auction. The second reflects the effect of an award on backlog, paid experience, and future eligibility. Equation (25) is the link between the model's dynamic mechanism and observed bids (Jofre-Bonet and Pesendorfer, 2003; De Silva et al., 2023).

Equilibrium A Markov perfect equilibrium would require each firm to track the state of every active rival in each period; the large number of active firms (almost 400) makes this solution concept computationally infeasible. Following De Silva et al. (2023), I instead use the moment-based Markov equilibrium of Ifrach and Weintraub (2017), which builds on related large-market solution concepts in Weintraub et al. (2008) and Weintraub et al. (2010), which approximates the Markov perfect equilibrium. Firms condition on the exact entrant set and rivals' public states in a current auction, but use $\bar{s}_t$ to forecast future competition.

Definition 1 (Moment-based equilibrium)**.** *Fix the exogenous policies: the RPPA category promotion kernel, the auction participation policy, and the entry and exit processes. A moment-based equilibrium in bids consists of type-symmetric bid functions and value functions, firm-specific type beliefs (within auctions), and a perceived law of motion for $\bar{s}_t$ held by all firms, such that:*

1. *in each auction, each firm that submits a bid maximizes (24), given the realized entrant set, rivals' public states, and beliefs over rival types;*
2. *value functions satisfy (22) under the fixed policies and the perceived law of motion;*
3. *beliefs over rival types follow Bayes' rule from the category-specific type shares and the participation policy, as in (17), and beliefs over rival bids are given by type mixtures (19) implied by equilibrium bid policies; and,*
4. *the perceived law of motion for the aggregate state $\bar{s}_t$ is consistent with the law induced by the fixed policies and firms' equilibrium behavior.*

5 Implementation and Estimation

In this section, I describe how I construct the estimation sample and estimate the exogenous policy environment. I first estimate the exogenous policy environment, permanent execution types, and

type-specific execution, participation, and bid policies. In the second stage, these objects determine auction hazards, static markups, and the transition policies used to compute continuation values. Because continuation values are computed from these first-stage objects alone and do not depend on the construction cost parameters, I solve the value function once and then estimate those parameters from the bidding first-order condition without re-solving firms' policies for each candidate parameter vector. This procedure follows the two-step approach to dynamic models in Hotz and Miller (1993); Bajari et al. (2007); Aguirregabiria and Mira (2007).

5.1 Sample Construction

I estimate the model on auctions conducted between September 2016 and October 2023. I define a firm as *regular* if it submits at least two positive, qualified bids on distinct lots before November 1, 2023. A firm enters the sample if it is regular.[13] This restriction yields a sample of 394 regular firms – the main sample used for analysis. For participation, a regular firm enters the risk set for a tender when it is eligible by category and has submitted a bid during the previous four quarters. The resulting panel contains 116,894 firm–tender pairs. Regular firms represent 56% of all bidders but 95% of all bids and 97% of all contracts won.

I restrict the auction sample to those in which at least one regular bidder competes, which yields 1,319 tender-lots and 3,035 bids. Among these, 460 auctions have a single qualified bidder. For rivals in the sample with an observed state, I evaluate the estimated bid policy at that state; for other rivals, I use the corresponding category and type-cell bid distribution (12.1% of bids). The project execution panel contains 13,440 firm–quarter observations for the 320 firms that win a contract (the remaining 74 firms win no contracts).

Both estimation stages use the firm and auction sample defined above. I first estimate completion, participation, and bidding policies, then use these policies to recover continuation values and construction costs. All monetary variables are expressed in nominal, VAT-exclusive Rwandan francs (RWF), rescaled to billions where indicated.

5.2 States and Policy Environment

Firm and Aggregate States In each quarter, I observe the category vector, cumulative awarded value W_{it}, and the current flow ΔW_{it} of Works contracts entering the portfolio. I construct backlog B_{it} recursively from these commitments and the VAT-implied Works payment flow Y_{it}^{W}, and define paid experience as $P_{it} = W_{it} - B_{it}$.[14] Quarterly commitments, backlog, and cumulative awarded value are constructed from signed contract values and Works payments (from VAT data). By construction, P_{it} is also cumulative allocated Works payment.[15]

[13] A lead bid is submitted by the firm itself rather than through a Joint Venture (JV). I exclude bids disqualified for administrative, technical, or financial reasons.

[14] I initialize backlog using the estimated scheduled remaining value of contracts active at the beginning of the panel and omit the first four quarters as a burn-in period. The ZOIB specification is estimated from the realized completion histories and propagates backlog in the value-function calculations and counterfactual simulations.

[15] Appendix C.1 gives the exact construction and its reconciliation with the payment data.

The participation, bid, and promotion policies use versions normalized by the scale of the firm's category. For firm i in category $\kappa_{im,t}$ in sector m at time t, I denote

$$\widetilde{B}_{it}(\kappa) = \frac{B_{it}}{S(\kappa)}, \qquad \widetilde{P}_{it}(m,\kappa) = \frac{P_{it}}{T_{m,\kappa}}, \tag{26}$$

where $S(\kappa)$ is the category capacity scale, measured as the median engineering estimate of category-κ tenders in all sectors, and $T_{m,\kappa}$ is the sector-category payment threshold. The numerators are firm totals, while the denominators vary with the firm's category.

The aggregate state $\bar{s}_t$ is defined in (4). Appendix C.4.3 details its construction, discretization, and estimated law of motion.

Industry Entry and Exit Firm entry is marked when the firm submits its first bid. At each period, the set of potential entrants consists of firms that registered but have not submitted a bid. Among the potential entrants, the quarterly entry hazard is

$$h_t^E = \Lambda\big(\gamma_{E0} + \bar{s}_t'\gamma_E\big), \qquad \text{where} \qquad \Lambda(u) = \frac{\exp(u)}{1+\exp(u)}. \tag{27}$$

Entrants' initial states and fixed characteristics are drawn from their empirical joint distribution conditional on the aggregate state. Conditional on $\bar{s}_t$, all potential entrants share the same hazard, so the number entering in quarter t is Binomial in the number of potential entrants.

A share of active firms exit the market each period. Among active firms, the quarterly exit hazard is

$$\zeta_{it} = \Lambda\big(\gamma_{X0} + g_X(s_{it})'\gamma_X + \bar{s}_t'\gamma_{Xs}\big), \tag{28}$$

where g_X contains category indicators, $\log(1+B_{it})$, and $\log(1+P_{it})$. Exit is absorbing. In the implementation, a firm is active from its first to its last observed bid and exits in the quarter of that last bid, with the last four sample quarters treated as right-censored so that every assigned exit is followed by at least four quarters without a bid.

Entry and exit are modelled as exogenous transitions rather than choices made by firms, so these hazards are held fixed in counterfactuals. Appendix C.4.2 reports the estimated hazards.

RPPA Category Transitions I parameterize the transition of each sector component $\kappa_{im,t}$ of the joint category kernel in (13) using a promotion hazard and a destination distribution. For an active sector m and current category $\kappa = \kappa_{im,t}$, the policy evaluates the firm-level paid experience stock against the requirements of that sector-category. After project execution realized in quarter $t-1$, the probability that a firm is promoted category in that sector is

$$h_{m\kappa}^R(P_{it}) = \Lambda\left[\alpha_{m\kappa} + \delta_0\,\mathbf{1}\{P_{it}>0\} + \beta_P \log\left(1+\frac{P_{i,t}}{T_{m,\kappa}}\right)\right]. \tag{29}$$

This probability increases smoothly with paid experience. Conditional on promotion, the destination rule assigns probability

$$p_m^R(\kappa' \mid \kappa, P_{i,t+1}), \qquad R(\kappa') > R(\kappa), \tag{30}$$

to each feasible higher category. Firms moving from E-B are restricted to upwards changes of at most two ranks; moves from F are unrestricted. Moving to category A is an absorbing state. Transitions into a previously inactive sector are included in the same category transition policy. Note that while paid experience is not reset after promotion, its normalization does change with the destination category, and the backlog state is also rescaled by $S(\kappa)/S(\kappa')$ at a category transition. When computing the value of winning, a firm applies these sector-specific transitions to the full category vector $\boldsymbol{\kappa}_{it}$.

5.3 First Stage: Execution Types and First-Step Policies

I assume each firm is one of two permanent latent types, H or L. I assume that firms' execution, auction participation, and bids depend on their type.

Execution For firms with outstanding work $B_{it} + \Delta W_{it} = X_{it} > 0$, I parametrize the distribution of work completed in (8) as a zero–one-inflated Beta distribution:

$$\begin{aligned}
\mathbb{P}(\xi_{it} = 0 \mid X_{it}, \theta_i) &= p_{0\theta_i}(X_{it}), \\
\mathbb{P}(\xi_{it} = 1 \mid \xi_{it} > 0, X_{it}, \theta_i) &= p_{1\theta_i}(X_{it}), \\
\xi_{it} \mid 0 < \xi_{it} < 1, X_{it}, \theta_i &\sim \text{Beta}\Big(\mu^{\xi}_{\theta_i}(X_{it})\phi, [1 - \mu^{\xi}_{\theta_i}(X_{it})]\phi\Big),
\end{aligned} \tag{31}$$

The Beta precision $\phi > 0$ is common across types. I implement the three cases as,

$$\begin{aligned}
\text{logit}\, p_{0\theta}(X_{it}) &= \alpha_0^{\xi} + \beta_0^{\xi} x_{it}^{\xi} - \lambda^{\xi} \mathbf{1}\{\theta = H\}, \\
\text{logit}\, p_{1\theta}(X_{it}) &= \alpha_1^{\xi} + \beta_1^{\xi} x_{it}^{\xi} + \lambda^{\xi} \mathbf{1}\{\theta = H\}, \\
\text{logit}\, \mu^{\xi}_{\theta}(X_{it}) &= \alpha_{\mu}^{\xi} + \beta_{\mu}^{\xi} x_{it}^{\xi} + \lambda^{\xi} \mathbf{1}\{\theta = H\}, \qquad \lambda^{\xi} > 0,
\end{aligned} \tag{32}$$

where $x_{it}^{\xi} = [\log X_{it} - \overline{\log X}]/\text{sd}(\log X)$ is the normalized outstanding work. The coefficients β_0^{ξ}, β_1^{ξ}, and β_{μ}^{ξ} allow the completion shares to vary with the post-award stock X_{it}. Execution type enters through the parameter $\lambda^{\xi} > 0$, which lowers the zero-completion probability and raises both the full-completion probability and the interior mean. This parameter restriction implements the stochastic ordering in (9). [16]

Participation For project ℓ, set $m = m_\ell$ and let $\Delta R_{i\ell t} = R(\kappa_{im,t}) - R(k_\ell)$ denote the category-rank gap, positive when the project lies below the firm's category. With type L as the omitted

[16] I also estimated a two-type Bernoulli specification based on indicators about whether quarterly payments met scheduled progress. The main qualitative results were largely robust to this alternative execution process.

category, the participation policy is

$$\begin{aligned}\mathbb{P}(a_{i\ell t}=1 \mid \theta_i) = \Lambda\Big(&\delta_A + \underbrace{\delta_M \mathbf{1}\{\kappa_{im,t}=k_\ell\} + \delta_{\bar{M}}\overline{\mathbf{1}\{\kappa_{im,t}=k_\ell\}}_i}_{\text{same category}} + \underbrace{\delta_R \Delta R_{i\ell t} + \delta_{\bar{R}}\overline{\Delta R}_i}_{\text{category-rank gap}} \\ &+ \underbrace{\delta_B \widetilde{B}_{it}(\kappa_{im,t}) + \delta_{\bar{B}}\overline{\widetilde{B}}_i}_{\text{backlog}} + \underbrace{\delta_P \log[1+\widetilde{P}_{it}(m,\kappa_{im,t})] + \delta_{\bar{P}}\overline{\log(1+\widetilde{P})}_i}_{\text{paid experience}} \\ &+ \delta_s' \bar{s}_t + \delta_H \mathbf{1}\{\theta_i = H\}\Big). \end{aligned} \tag{33}$$

I control for firm-specific mean terms (denoted with bars) to separate changes in a firm's state from persistent differences across firms (Mundlak, 1978).

Bids Let $\widetilde{b}_{i\ell t} = b_{i\ell t}/z_\ell$ be firm i's bid in project l at time t divided by the engineering cost estimate of project l. Conditional on firm i's private type θ_i, this normalized bid is assumed to be drawn from a type-specific Gamma distribution,

$$\widetilde{b}_{i\ell t} \mid \theta_i \sim \text{Gamma}\left(\nu_{\theta_i}, \frac{\nu_{\theta_i}}{\mu^b_{\theta_i}(\chi_\ell, s_{it}, \bar{s}_t)}\right), \tag{34}$$

where $\nu_\theta > 0$ and the second argument is the rate, so μ^b_θ is the conditional mean which depends on project-level covariates, the firm's individual state, and the aggregate state. For a vector $x^b_{i\ell t}$ containing the standardized normalized states (26), project scale, bidder count, calendar quarter, sector indicators, and the aggregate state, I set

$$\log \mu^b_\theta(\chi_\ell, s_{it}, \bar{s}_t) = \alpha^b_{\theta,\kappa_{im,t},k_\ell} + (x^b_{i\ell t})'\beta^b_\theta, \qquad m = m_\ell. \tag{35}$$

The intercepts vary by firm-category–tender-category cell. These bid policies recover the type-specific densities and survival functions used in (19).

Firm Type Likelihood For firm i, let $\mathcal{D}^\xi_i$, $\mathcal{D}^b_i$, and $\mathcal{D}^a_i$ denote its execution, normalized-bid, and participation histories. Conditional on type and the covariates in each component, the firm likelihood is

$$\mathcal{L}_i = \sum_{\theta\in\{L,H\}} \omega_\theta \, \mathcal{L}^\xi_i(\theta)\mathcal{L}^a_i(\theta)\mathcal{L}^b_i(\theta). \tag{36}$$

Appendix C.5.1 derives this factorization from the chain rule and the maintained Markov and conditional independence restrictions.

Here $\omega_\theta = \mathbb{P}(\theta_i = \theta)$, with $\omega_H = \Lambda(\alpha^\omega)$ and $\omega_L = 1 - \omega_H$. The firm's type likelihood depends on the product of three likelihoods: for project execution, auction participation, and bidding. Each block is a product over that firm's own observations. The execution block depends on the product

of the three cases of (31):

$$\mathcal{L}_i^{\xi}(\theta) = \prod_{t \in \mathcal{D}_i^{\xi}} \Big[p_{0\theta}(X_{it})\Big]^{\mathbf{1}\{\xi_{it}=0\}} \Big[\big(1-p_{0\theta}(X_{it})\big)p_{1\theta}(X_{it})\Big]^{\mathbf{1}\{\xi_{it}=1\}} \times \Big[\big(1-p_{0\theta}(X_{it})\big)\big(1-p_{1\theta}(X_{it})\big)\, f_{\mathrm{B}}\big(\xi_{it};\, \mu_\theta^{\xi}(X_{it})\phi,\, [1-\mu_\theta^{\xi}(X_{it})]\phi\big)\Big]^{\mathbf{1}\{0<\xi_{it}<1\}}, \tag{37}$$

where f_{B} is the Beta density, and the three components are as in (32). Letting $\eta_{i\ell t}(\theta)$ denote the index inside $\Lambda(\cdot)$ in (33), which differs across types only through the level shift δ_H, the participation block is

$$\mathcal{L}_i^{a}(\theta) = \prod_{(\ell,t)\in\mathcal{D}_i^{a}} \Lambda\big(\eta_{i\ell t}(\theta)\big)^{a_{i\ell t}} \Big[1-\Lambda\big(\eta_{i\ell t}(\theta)\big)\Big]^{1-a_{i\ell t}}. \tag{38}$$

With $\widetilde{b}_{i\ell t} = b_{i\ell t}/z_\ell$ and μ_θ^b from (35), the Gamma density in (34) gives the bid block

$$\mathcal{L}_i^{b}(\theta) = \prod_{(\ell,t)\in\mathcal{D}_i^{b}} \frac{1}{\Gamma(\nu_\theta)} \left(\frac{\nu_\theta}{\mu_\theta^b(\chi_\ell, s_{it}, \bar{s}_t)}\right)^{\nu_\theta} \widetilde{b}_{i\ell t}^{\,\nu_\theta - 1} \exp\left(-\frac{\nu_\theta\, \widetilde{b}_{i\ell t}}{\mu_\theta^b(\chi_\ell, s_{it}, \bar{s}_t)}\right). \tag{39}$$

Bids enter conditional on participation, so (38) and (39) together factor the joint distribution of the participation decision and the bid it produces.

EM algorithm Instead of maximizing (36) directly, I use the Expectation–Maximization algorithm in Arcidiacono and Miller (2011). It divides that procedure into two simpler steps, iterated to convergence. The E-step fills in the missing type, computing for each firm the posterior probability that it is type H given its complete history. The M-step then proceeds as if types were observed, re-estimating each policy on a sample in which every observation appears once as each type, weighted by that posterior. Assume iteration r has occurred; then, iteration $r+1$ proceeds as follows.

1. **Expectation step.** Let

$$\Psi = (\alpha^{\omega}, \psi^{\xi}, \psi^{b}, \psi^{a})$$

collect the parameters for the type share and the execution process, normalized-bid, and participation policies. Given $\Psi^{(r)}$, compute the posterior $\widehat{w}_i^{(r)}$ for each firm. Because $\omega_H = \Lambda(\alpha^{\omega})$, this takes a log-odds form,

$$\operatorname{logit} \widehat{w}_i^{(r)} = \alpha^{\omega(r)} + \sum_{c\in\{\xi,b,a\}} \Big[\ln \mathcal{L}_i^{c}\big(H; \psi^{c(r)}\big) - \ln \mathcal{L}_i^{c}\big(L; \psi^{c(r)}\big)\Big], \tag{40}$$

so each block enters the classification of firm i as an additive log-likelihood ratio, and a firm is classified H when its execution, bidding, and participation records jointly favor H over L.

2. **Maximization step.** Given $\{\widehat{w}_i^{(r)}\}$, choose $\Psi^{(r+1)}$ to maximize

$$Q\big(\Psi \mid \Psi^{(r)}\big) = \sum_{i=1}^{N} \sum_{\theta \in \{L,H\}} w_{i\theta}^{(r)} \Big[\ln \omega_\theta + \ln \mathcal{L}_i^{\xi}(\theta) + \ln \mathcal{L}_i^{b}(\theta) + \ln \mathcal{L}_i^{a}(\theta) \Big], \tag{41}$$

with $w_{iH}^{(r)} = \widehat{w}_i^{(r)}$ and $w_{iL}^{(r)} = 1 - \widehat{w}_i^{(r)}$. Because no parameter enters more than one likelihood block, (41) separates into four independent maximization problems, which are solved in sequence (Arcidiacono and Jones, 2003):

(a) the type share in closed form, $\omega_H^{(r+1)} = N^{-1} \sum_i \widehat{w}_i^{(r)}$;

(b) the execution parameters ψ^{ξ} in (32), by weighted maximum likelihood;

(c) the bid policy ψ^{b} in (35), by weighted maximum likelihood within each firm-category–tender-category cell;

(d) the participation policy ψ^{a} in (33), by weighted logit on the eligible risk set.

In (b)–(d) each observation enters twice, once as type L with weight $1 - \widehat{w}_i^{(r)}$ and once as type H with weight $\widehat{w}_i^{(r)}$.

Posteriors are updated with a damping factor of 0.5, and iteration stops when the sup-norm change is below 10^{-3} in the posteriors and 2×10^{-3} in the stacked policy vector.[17]

5.4 Second Stage: Auction Hazards, Continuation Values, and Costs

The first stage provides the type posteriors and the type-specific completion, participation, and bidding policies. In the second stage, I use these estimates to construct auction markups, calculate how winning changes a firm's continuation value, and estimate construction costs. I hold the estimated policies fixed throughout this stage, rather than re-solving participation and bidding for each candidate cost parameter.

Auction hazards and markups. For each firm category, I calculate the prior probability that a firm is high-type using the estimated type posteriors of firms in that category. Observing that a rival participates provides additional information about its type. I therefore update this prior using the type-specific participation probabilities in (17). I evaluate both types' Gamma bid distributions at the rival's public state and combine their densities and survival functions using these updated probabilities, as in (19).

The resulting estimated auction markup is denoted

$$\widehat{\mu}_{i\ell t} = \frac{1}{\widehat{H}_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})}. \tag{42}$$

[17] Any $\Psi^{(r+1)}$ that raises $Q(\Psi \mid \Psi^{(r)})$ above $Q(\Psi^{(r)} \mid \Psi^{(r)})$ also raises the observed log likelihood (36), so each pass weakly improves the fit.

Rival identities are used to recover public states and qualification categories, but not realized types or the econometrician's posterior.

Construction cost specification. I parameterize construction costs relative to the engineering estimate as

$$\begin{aligned}\frac{c_{i\ell t}}{z_\ell} = \underbrace{\alpha^{\text{tend}}_{k_\ell} + \alpha^{\text{firm}}_{\kappa_{im_\ell,t}} + \alpha^{\text{sec}}_{m_\ell}}_{\text{category and sector effects}} + \underbrace{\gamma\frac{\log z_\ell}{25} + q_\ell'\beta_c}_{\text{project characteristics}} \\ + \underbrace{\rho_B \log(1+B_{it}) + \rho_P \log(1+P_{it}) + \rho_{BP}\log(1+B_{it})\log(1+P_{it})}_{\text{backlog and paid experience}} \\ + \underbrace{\eta_H \mathbf{1}\{\theta_i = H\}}_{\text{execution type}} + \varepsilon_{i\ell t},\end{aligned} \tag{43}$$

where backlog and paid experience are measured in billions of Rwandan francs. The vector q_ℓ contains BOQ item count, advertised duration, procuring entity type, the lagged material price shock, and missing-value indicators.

Let ϑ collect the coefficients that enter the conditional mean in (43), and write its covariates as $x_{i\ell t}(\theta)$. Conditional on public state and type, $\varepsilon_{i\ell t} \sim N(0, \sigma^2_{i\ell t})$ independently across firm–project matches. Residual dispersion in costs may depend on project size:

$$\log \sigma_{i\ell t} = \alpha^\sigma_0 + \beta^\sigma_z \left(\log z_\ell - \overline{\log z}\right). \tag{44}$$

In a static auction, costs could be recovered by subtracting the estimated markup (42) from firms' bids. In our setting, the bid also reflects the continuation value of winning, which I recover as follows.

Expected current profits. I compute continuation values under the participation and bid policies estimated in the first stage. For each type and point on the (discretized) state grid, I integrate over the tender categories for which the firm is eligible. The arrival intensity λ_k is set to the average number of category-k lots per quarter in the estimation window, and the characteristics distribution G^χ_k is approximated by a 12-point discretized lot distribution, which simulates representative projects and their characteristics from observed projects.[18]

The flow payoff at a state is the expected profit per quarter from bidding. By the bid first-order condition, a firm's profit from an auction at its optimal bid, inclusive of the continuation value of winning, equals its winning probability divided by the hazard (De Silva et al., 2023). Let

$$\pi^A_{i\ell t} = \frac{G_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})}{H_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})}$$

[18]Within each tender category, I sort observed projects by their engineering estimate, divide them into twelve approximately equal-sized strata, and use the median project in each stratum, weighted by the stratum's empirical frequency.

denote this profit for firm i in auction ℓ at time t. The flow payoff at state s is the expected sum of these profits over the auctions the firm bids on in a quarter,

$$\Pi_\theta^A(s) = E\left[\sum_{\ell \in \mathcal{O}_{it}} a_{i\ell t}\, \pi_{i\ell t}^A \;\middle|\; s_{it} = s,\ \theta_i = \theta\right], \tag{45}$$

where $\mathcal{O}_{it}$ is the set of lots arriving in quarter t for which i is eligible, and $a_{i\ell t}$ is the participation indicator. Conditional on the firm remaining in the market at the start of the quarter, the expectation averages over project arrivals, participation, the firm's bid, and the set of rivals under the estimated policies. Let $\mathbf{\Pi}_\theta^A$ denote the vector that stacks $\Pi_\theta^A(s)$ over the state grid.

For every bid, $\widehat{G}_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})/\widehat{H}_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})$ is the estimated winning probability divided by the estimated hazard at that bid. I regress this ratio on project characteristics, the firm's state, the aggregate state, and type. Each bid enters once for each type, weighted by the firm's posterior probability of that type. I evaluate the fitted regression at the representative projects and grid states, including states with no observed bids. Multiplying by participation probabilities and auction arrival rates and summing across categories gives $\mathbf{\Pi}_\theta^A$. Because the estimated participation and bidding policies remain fixed, the transition matrix does not depend on the cost parameters ϑ.

State transitions. Let M_θ^0 denote the no-award transition matrix for type θ. In a quarter without new awards, firms complete outstanding work, accumulate paid experience, and may be promoted. The aggregate state also changes, and firms may exit before the next quarter's auctions. The exit state has zero value and is omitted from the matrix, so each row sums to the probability of remaining in the market next quarter. I use the no-award transition because the auction surplus in (45) already includes the continuation gain from winning.

For each award, I construct the next-period state using the transitions in (10)–(13). The quarterly discount factor is set to $\beta = 0.95$. Following Jofre-Bonet and Pesendorfer (2003), I evaluate the infinite-horizon Bellman system under the participation and bidding behavior estimated in the first stage. Let $\mathbf{V}_\theta$ denote the vector obtained by evaluating the value function $V_\theta(s, \bar{s})$ at every point on the state grid. Because the estimated policies and the induced transition matrix are held fixed, this vector is

$$\mathbf{V}_\theta = (I - \beta M_\theta^0)^{-1} \mathbf{\Pi}_\theta^A. \tag{46}$$

This is the auction-surplus representation of De Silva et al. (2023). Appendix D.3 shows its equivalence to the numerical implementation, which stores values before the current exit draw. Because neither $\mathbf{\Pi}_\theta^A$ nor M_θ^0 depends on the cost parameters, the Bellman system is solved once per type, and the construction cost parameters are recovered only afterwards. Evaluating (46) solves an infinite-horizon Bellman system.[19]

I compute the dynamic values for each $\theta \in \{L, H\}$ separately. Thus, the flow payoffs, transition

[19] For numerical validation, I also compute continuation values by simulating the same policy and transition law forward for five hundred quarters. The results match.

matrix, and value vector—$\mathbf{\Pi}_\theta^A$, M_θ^0, and $\mathbf{V}_\theta$—are type-specific because execution, participation, and bidding policies differ by type.

Continuation value of the focal award. For each observed bid, I create separate rows, $Q^{\text{win}}_{\theta,i\ell t}$ and $Q^{\text{lose}}_{\theta,i\ell t}$, that represent the path if the firm wins and loses the focal award, respectively. I construct both paths using the model's state transitions, with an award increment of z_ℓ after a win and zero after a loss. Both integrate over the estimated completion, promotion, aggregate-state, and exit transitions. As with M_θ^0, the zero-valued exit state is omitted from these rows. The dynamic gain entering the bid first-order condition is the discounted change in next-quarter continuation value,

$$\Delta V_{\theta,i\ell t} = \left(Q^{\text{win}}_{\theta,i\ell t} - Q^{\text{lose}}_{\theta,i\ell t}\right)\mathbf{V}_\theta, \qquad D_{\theta,i\ell t} = \beta\,\Delta V_{\theta,i\ell t}. \tag{47}$$

The current winning uses the award value of focal project ℓ; the continuation paths integrate over future tender arrivals using representative project values.

Structural estimating equation. Substituting (43), (42), and (47) into the bid first-order condition gives

$$\frac{b_{i\ell t} - \widehat{\mu}_{i\ell t} + D_{\theta,i\ell t}}{z_\ell} = x_{i\ell t}(\theta)'\vartheta + \varepsilon_{i\ell t}, \quad \text{where,} \quad \widehat{\mu}_{i\ell t} = \frac{z_\ell}{\widehat{H}_i(b_{i\ell t}/z_\ell)}. \tag{48}$$

Because $D_{\theta,i\ell t}$ is computed from the value vector before construction cost parameters are estimated, the left-hand side is data once the first stage and the value solve are complete, and ϑ enters only through the cost specification (43) on the right-hand side. Thus the observed bid is separated into construction cost, a static markup, and a dynamic markdown. Each bid contributes one type-L and one type-H row to (48), weighted respectively by $1-\widehat{w}_i$ and $\widehat{w}_i$. I estimate ϑ by the posterior-weighted heteroskedastic likelihood implied by (44). Firms know their own types, while the econometrician integrates over type using the posterior estimated in the first stage.

5.5 Inference

Inference uses 300 firm-level bootstrap samples. I hold the first stage fixed and re-estimate every second-stage object in each sample, so the reported intervals for continuation values and costs reflect uncertainty in the set of firms, their costs, and the continuation values, conditional on the estimated types and policies. I construct confidence intervals for estimated parameters and the dynamic value of winning ($\bar{D}_L$, $\bar{D}_H$, $\bar{D}_H - \bar{D}_L$) from the draw-level estimates.

6 Identification Discussion

Identification proceeds in three steps. First, the administrative histories identify the policy functions and state transitions used in the model. Second, repeated execution and auction histories identify the distribution of permanent execution types and the type-specific bid, participation, and project

execution policies. Third, the auction first-order condition identifies the construction cost net of the continuation value of winning. Construction costs and continuation values are then separated using the dynamic value and linear structure implied by the model.

State Transitions and Policy Functions At the beginning of quarter t, I observe the category vector $\boldsymbol{\kappa}_{it}$ and cumulative awarded Works value W_{it}. During the quarter, I observe the value ΔW_{it} of Works contracts entering the firm's portfolio and construct the Works payment flow Y_{it}^{W} from the VAT records. I then construct backlog recursively using

$$W_{i,t+1} = W_{it} + \Delta W_{it},$$
$$B_{i,t+1} = B_{it} + \Delta W_{it} - Y_{it}^{W}.$$

The payment flow is capped by the portfolio available to receive payment, so $B_{i,t+1}$ is nonnegative. I treat the resulting measured state $s_{it} = (\boldsymbol{\kappa}_{it}, B_{it}, W_{it})$, and the aggregate state $\bar{s}_t$ (which is observed), as given when estimating the policy functions.

Tender arrivals, industry entry and exit, and the aggregate state transition are assumed to be exogenous transition policies that are observed from their corresponding panels. The promotion policy is identified from category transitions conditional on origin category, sector, and paid experience, all of which are observed in each period. As in De Silva et al. (2023), auction participation is treated as a descriptive matching policy. I observe all eligible active firms and all their bid submissions to all tenders, so the participation policy is identified from match rates by eligible firm-tender pairs. These objects are identified (on the relevant support) under the maintained Markov restriction that, conditional on $(s_{it}, \bar{s}_t)$ and the current tender, earlier history does not directly affect the corresponding policy or transition.

Permanent Execution Types Identification of θ_i relies on firm-level repeated histories, which distinguish permanent heterogeneity from transitory shocks. Specifically, Kasahara and Shimotsu (2009) show that a model with a finite number of permanent types can be identified if current outcomes satisfy a Markov restriction, if there is sufficient variation in observed states that affects types differentially, and if the type-specific choice probabilities satisfy a rank condition ensuring that the types' response profiles are linearly independent. In my context, the Markov restriction is that conditional on $(\theta_i, s_{it}, \bar{s}_t, \chi_\ell)$, the per-period draws generating completion shares, participation, and the conditional bid are independent, and draws are independent across quarters and auctions. Thus, the only dependence over time arises from the firm's individual state and the common aggregate state.

Intuitively, identification is based on observing repeated sequences of firm behavior that persistently differ between the two types. For example, consider two firms facing the same individual and aggregate states. A single quarter of fast execution could reflect a favorable transitory draw. But, if one firm repeatedly completes a larger share of its outstanding work, participates according to the type-H policy, and submits bids drawn from the type-H distribution, that sequence is more likely

to have been generated by $\theta_i = H$ than by $\theta_i = L$. The conditions in Kasahara and Shimotsu (2009) ensure that a finite number of groups of firm types can be distinguished and thus the underlying finite-mixture can be identified.

I assume that execution type is a permanent and immutable firm characteristic; firms cannot choose to exert effort to increase their execution speed after being awarded a contract. The model thus rules out moral hazard. Any persistent effort effect is attributed to θ_i.

Auction Hazards and Cost Inversion Conditional on participation, the first-stage finite-mixture model identifies the parameters of the type-specific bid distribution

$$F_\theta^b\left(\widetilde{b} \mid \chi_\ell, s_{it}, \bar{s}_t\right), \qquad \widetilde{b}_{i\ell t} = \frac{b_{i\ell t}}{z_\ell},$$

under the parametric Gamma specification in (34)–(35), and identification of the type.

For each auction, I observe the participating bidder set and each bidder's public state. Bidders form beliefs about rival types by updating a category-specific prior over rival types using the type-specific probability of participation. Combining this probability with the type-specific bid distributions identifies each rival's bid density and survival function. These objects identify the auction hazard $H_i(b \mid \mathcal{J}_{\ell t})$ on the support of observed bids (Athey and Haile, 2007). The static markup $1/H_i(b \mid \mathcal{J}_{\ell t})$ is well-defined on every row because the bidder's hazard is a sum of Gamma hazards with shapes above one (the smallest being 6.5). For single-bidder lots, the hazard comes from the simulated rival.

Given the discount factor, the estimated bidding and participation policies determine auction surplus, while the estimated state transitions determine the no-award transition matrix. Together, these objects determine the value function through (46), without requiring the construction cost parameters. Winning differs from losing because an award increases backlog and cumulative awarded value. These changes affect subsequent payments, participation, promotion, and eligibility. The difference between the winning and losing value functions defines the continuation value $D_{i\ell t}$, which does not depend on the unknown cost parameters conditional on the first-step policies (Hotz and Miller, 1993; De Silva et al., 2023).

Given the auction hazard and continuation value, construction cost is recovered bid by bid using the Guerre et al. (2000)-style inversion adapted to the dynamic auction setting, as in Jofre-Bonet and Pesendorfer (2003):

$$b_{i\ell t} - \frac{1}{H_i(b_{i\ell t} \mid \mathcal{J}_{\ell t})} = c_{i\ell t} - D_{\theta_i, i\ell t}.$$

On the left-hand side, bids are observed and the auction hazard, and therefore the static markup, is identified as described above. Thus, the auction inversion identifies project cost net of the continuation gain from winning, but does not identify the two components separately.

Separating Costs and Continuation Values The auction inversion identifies only the net cost $c_{i\ell t} - D_{\theta,i\ell t}$. Because the continuation gain $D_{\theta,i\ell t}$ in (47) is computed from the first-stage policies and the value vector alone, the net cost separates into its two components bid by bid, without reference to the cost parameters. The construction cost specification (43) then models the recovered cost as $x_{i\ell t}(\theta)'\vartheta + \varepsilon_{i\ell t}$, and Equation (48) identifies ϑ given the other observed or estimated components.

Conditional on the first-step objects, and under the model's cost specification, the cost parameters ϑ are identified in our model under the key assumption that

$$\varepsilon_{i\ell t} = \sigma_{i\ell t} u_{i\ell t}, \qquad u_{i\ell t} \overset{\text{iid}}{\sim} N(0,1),$$

conditional on tender characteristics, public state, and type, where $\sigma_{i\ell t}$ is specified in (44). That is, the structural cost errors are conditionally independent across firm–project matches, firms, and periods. It is also necessary that the cost variables in (48) have full column rank.

Intuitively, the same variables enter current costs and continuation values in different ways. For example, current backlog affects the cost of executing the focal project through its level at the time of bidding. Its effect on the continuation value instead depends on how winning changes the firm's future backlog relative to losing, which varies with project size, execution type, paid experience, and the probability of promotion. As a result, firms with similar current values in the cost covariates can face different future effects from winning. This variation separates current construction cost from the continuation value (De Silva et al., 2023).

Counterfactuals The counterfactuals are identified conditional on four restrictions. First, the project cost distribution and type-specific completion-share functions remain fixed. Second, the estimated tender arrival, participation, entry, and exit processes, as well as the category promotion policy, remain the same under the counterfactual rule except where the exercise explicitly changes them. Third, firms use the observed public state and current rival states in each auction, while forecasting future competition from $\bar{s}_t$. Fourth, the counterfactual remains within the support over which the type-specific policies and cost distribution are estimated. Thus, the counterfactual changes the contract award mechanism, competition in auctions, backlog, paid experience, and dynamic markdowns, but holds the environment, government behavior, and project completion speeds fixed.

7 Estimation Results

This section presents the structural model estimates, quantifies the magnitude of the dynamic incentives, and the effectiveness of screening in this market. Section 7.1 presents the results of Stage 1: execution types and the estimated participation and bid policies; Section 7.2 reports the results of Stage 2 cost and continuation-value estimates, and decomposes bids to show how dynamic incentives induce high-type firms to shade their bids; Appendix E.1 assesses model fit.

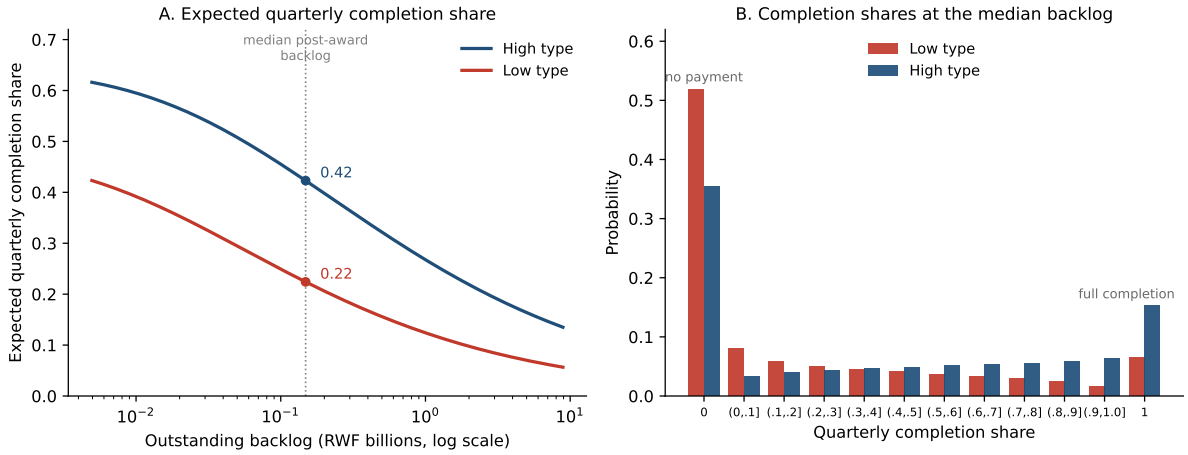
7.1 First-stage estimates: firm types and participation and bid policies

The EM algorithm estimates type-specific distributions of quarterly completion shares that differ sharply between high- and low-type firms. The algorithm assigns 19.8 percent of firms to the high type; the remaining 80.2 percent is assigned to the low type.

The estimated completion process differs across types on each of its three margins. At the median backlog stock, high-type firms have zero payment in 36 percent of quarters, compared with 52 percent for low-type firms. They complete all outstanding work in 15 percent of quarters, compared with 7 percent for low-type firms. Among quarters with partial completion, the high type completes 55 percent of its portfolio on average, compared with 39 percent for the low types. Figure 6, Panel B, plots the implied distribution of quarterly completion shares at the median backlog. Combining these margins, the high type completes 42.3 percent of its outstanding portfolio per quarter ($[33.1, 49.9]$ percent), compared with 22.4 percent for the low type ($[18.9, 26.3]$ percent). This implies that, with 148.5 million RWF of backlog, a high-type firm finishes half of their outstanding work in 1.3 quarters and 90 percent in 4.2 quarters, while a low-type firm takes approximately 2.7 and 9.1 quarters, respectively.

The distribution of quarterly completion shares varies with the amount of work outstanding. As the post-award stock increases, the probability of completing all outstanding work and the share completed in partial-payment quarters fall, while the probability of zero completion changes little. Larger portfolios therefore have lower quarterly completion shares for both types (Figure 6, Panel A). Appendix Table 14 reports the estimated zero one inflated Beta parameters and confidence intervals.

Figure 6: Estimated quarterly completion shares by type



Notes: Panel A: expected quarterly completion share as a function of the backlog stock. The dotted line marks the median post-award backlog: 148.5 million RWF. Panel B: the implied distribution of quarterly completion shares at the median backlog. The bar at 0 represents no payment, and the one at 1 represents completion of all outstanding work. The interior bars are the Beta component integrated over deciles. Parameters are the point estimates reported in Appendix Table 14.

Table 1 reports the main first-stage estimates for the execution, participation, and bid policies. The participation policy includes each firm's mean backlog and paid experience, so the coefficients on the current states are identified from deviations from those means. Firms that operate with greater

Table 1: First-stage estimates

Quantity	Estimate	95% bootstrap CI
Execution and type		
High-type population share (ω_H)	19.8%	[13.4%, 27.5%]
Expected quarterly completion share, L ($\mathbb{E}[\xi \mid X_{\text{med}}, L]$)	22.4%	[18.9%, 26.3%]
Expected quarterly completion share, H ($\mathbb{E}[\xi \mid X_{\text{med}}, H]$)	42.3%	[33.1%, 49.9%]
Qualified participation		
Current backlog (δ_B)	−0.242	[−0.339, −0.163]
Paid experience (δ_P)	−0.488	[−0.650, −0.337]
Same-category tender (δ_M)	−0.728	[−1.030, −0.361]
Category gap (δ_R)	−1.204	[−1.431, −0.947]
Firm mean backlog ($\delta_{\bar{B}}$)	0.454	[0.285, 0.691]
Firm mean paid experience ($\delta_{\bar{P}}$)	0.617	[0.349, 0.911]
High type (δ_H)	1.331	[1.054, 1.593]
Conditional normalized-bid policy		
Gamma shape, L (ν_L)	6.84	[6.06, 7.96]
Gamma shape, H (ν_H)	15.44	[12.62, 22.44]
Mean predicted bid / engineering estimate, L ($\overline{\mu}^b_L$)	1.055	[1.027, 1.091]
Mean predicted bid / engineering estimate, H ($\overline{\mu}^b_H$)	1.038	[1.011, 1.079]
Mean predicted bid difference, H minus L ($\overline{\mu}^b_H - \overline{\mu}^b_L$)	−0.017	[−0.068, 0.030]

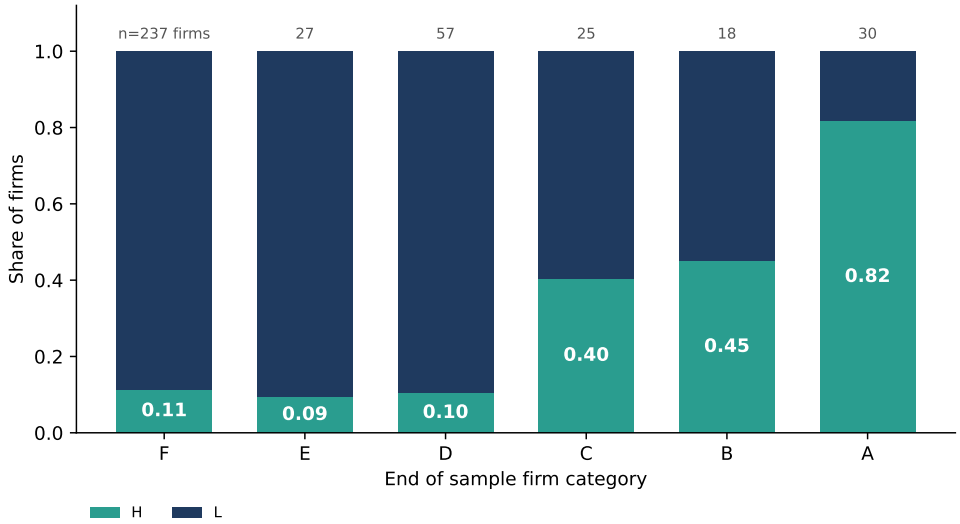
Notes: First-stage execution, participation and bid-policy point estimates. Intervals are 95 percent percentile intervals from 300 firm-bootstrap draws that re-estimate the first step (execution, participation and bid policy) on each resampled firm set, with the EM started from the point's type posteriors. I evaluate both types' Gamma bid policies at the same observed states and auction cells. The H-minus-L difference compares predicted bids at these common states. $\overline{\mu}^b_\theta$ denotes the sample average of the conditional mean μ^b_θ. Additional parameter estimates are reported in the appendix.

backlog and with more paid experience participate more on average. However, firm participates less in quarters when its backlog or paid experience is above its average level. High-type firms participate more often conditional on the same state.

Comparing firms at the same state and in the same auction, high- and low-type firms have similar predicted bids. The difference is small and not statistically significant. The differences between types are more pronounced in completion and participation.

Figure 7 shows that high-type firms are concentrated at the top of the category ladder. At the end of the sample, they account for 82 percent of firms in category A and 45 percent in category B, compared with 11 percent in category F. Each firm is assigned to its highest registration category across sectors.

Figure 7: Execution types by registration category



Notes: Firms are assigned to their highest registration category across sectors in the final quarter of the sample. High-type shares are means of the estimated posterior probabilities, weighting firms equally. Numbers above the bars report the number of firms in each category. The sample contains 394 firms.

7.2 Second-stage estimates: cost and continuation value

Table 2 reports the main parameter estimates recovered in the second stage. High-type firms have higher estimated costs than low-type firms. The full-sample estimate of the high-type cost coefficient η_H is 0.437 of the ECE, or 43.7 percentage points (bootstrap CI $[-0.207, 1.085]$). The coefficients on backlog, paid experience, and their interaction are -0.104, -0.115, and 0.017; none is statistically different from zero, with 95 percent bootstrap intervals of $[-0.635, 0.768]$, $[-0.462, 0.417]$ and $[-0.442, 0.260]$. Table 2 also reports the firm-category, tender-category and sector effects, the project-scale coefficient and the cost dispersion; the project controls are in Appendix E.

Using the estimated cost and continuation values, I decompose bids for high- vs. low-type firms for otherwise similar firms. Figure 8 presents the model-generated bid decomposition for firms in the same tender and public state. The high type's project cost is estimated to be 43.7pp higher (as a percent of the ECE), while its dynamic markdown is 37.7pp larger, so the dynamic channel offsets most of the cost gap. On average, the total bid is 6.0 points higher for a high-type firm, and it bids below its low-type counterpart in 48.6 percent of bids (holding the firms' states and tender characteristics fixed).

Table 3 measures the dynamic markdown, $D_{\theta_i,i\ell t}$, relative to the ECE. The high-type markdown is much larger than the low-type markdown in every category in percentage point terms. As a percent of the ECE, the markdown is smallest for category A firms, which have no prospect of promotion to a higher category. It is largest in the middle of the ladder, reaching 76 percent of the ECE for high-type firms in category E and ranging from 50–64 percent between categories D–B.

Table 2: Second-stage cost estimates

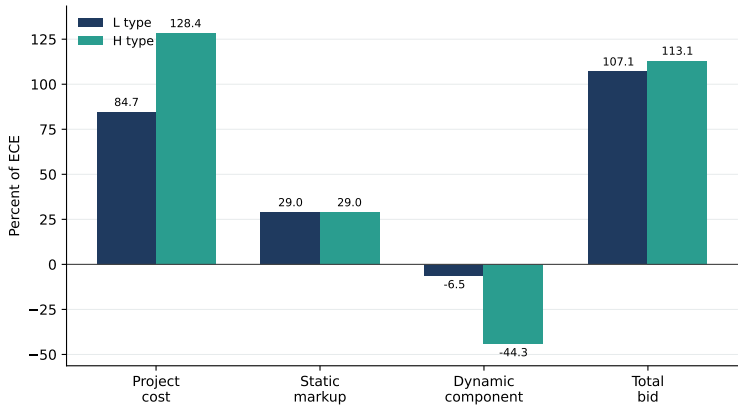
Quantity	Estimate	95% bootstrap CI
Current cost: type, states, and scale (share of the engineering estimate)		
High type (η_H)	0.437	[−0.207, 1.085]
Log backlog (ρ_B)	−0.104	[−0.635, 0.768]
Log paid experience (ρ_P)	−0.115	[−0.462, 0.417]
Backlog × paid experience (ρ_{BP})	0.017	[−0.442, 0.260]
Log project scale (γ)	−1.667	[−3.860, 0.583]
Firm-category effects (categories A and B pooled and omitted)		
Category F (α_F^{firm})	−0.495	[−1.082, 0.359]
Category E (α_E^{firm})	−0.474	[−1.114, 0.325]
Category D (α_D^{firm})	−0.291	[−0.826, 0.264]
Category C (α_C^{firm})	−0.185	[−0.626, 0.253]
Tender-category effects (equation intercepts)		
Category F (α_F^{tend})	2.633	[0.827, 4.375]
Category E (α_E^{tend})	2.605	[0.659, 4.349]
Category D (α_D^{tend})	2.629	[0.656, 4.424]
Category C (α_C^{tend})	2.606	[0.560, 4.509]
Category B (α_B^{tend})	2.189	[−0.126, 4.095]
Category A (α_A^{tend})	2.349	[0.244, 4.387]
Sector effects (Buildings omitted)		
Roads ($\alpha_{\text{Roads}}^{\text{sec}}$)	−0.018	[−0.591, 0.178]
Water ($\alpha_{\text{Water}}^{\text{sec}}$)	−0.205	[−0.445, 0.091]
Cost dispersion		
σ	0.607	[0.536, 1.091]
Log-project-scale variance slope (β_z^{σ})	−0.090	[−0.197, −0.002]

Notes: Second-stage estimates of the construction cost equation (43). The Estimate column reports full-sample point estimates; intervals are 95 percent percentile intervals from 300 firm-bootstrap draws that hold the first stage fixed. Cost coefficients are expressed as shares of the engineering cost estimate. Backlog and paid experience enter as $\log(1+\cdot)$ in billions of VAT-exclusive Rwandan francs; project scale enters as $\log z_\ell/25$. The tender-category terms are the intercepts of (43): the equation has no separate constant, so they are not deviations from an omitted tender category. The omitted firm category pools A and B, and the omitted sector is Buildings. Project controls (BOQ item count, advertised duration, procuring-entity type, the lagged material-price shock, and their missing-value indicators) are estimated and reported in Appendix E.

8 Counterfactual policies

I use the model to conduct counterfactuals to evaluate Rwanda's eligibility restrictions based on past performance. First, I evaluate the cost of screening to the government by loosening eligibility requirements and allowing firms from categories below the auction's current requirement to participate. Second, I change the RPPA category promotion rule to quantify how conditioning future access on completed work affects bidding incentives and contract allocation across firm types. As a benchmark, I compare the existing rule with random promotion within each category, which removes the link between completing a contract and the probability of being promoted. To understand how much more procurement outcomes could improve using past performance to screen firms, I then search over similar rules to minimize value-weighted delays. Together, these

Figure 8: Decomposing bids by firm type



Notes: Baseline specification, all 2,982 supported bids, unweighted means over paired type evaluations, in percent of the engineering estimate. Each type's bid is constructed from the fitted cost equation on the same tender and public state, holding the static markup at its observed-auction value, so rivals' bidding is unchanged. The figure compares otherwise identical firms that differ only in execution type.

Table 3: The dynamic component of bids, by bidder category and type

		High type		Low type	
Bidder category	Bids	Gain (M RWF)	% of ECE	Gain (M RWF)	% of ECE
A	215	324.0	12.4	22.6	1.6
B	93	505.9	64.2	30.4	3.2
C	531	174.1	51.1	8.6	2.5
D	636	104.6	49.8	8.5	8.9
E	210	56.8	75.7	3.0	6.5
F	1,297	9.1	38.8	1.9	8.4
All	2,982	107.2	46.7	2.5	7.7

Notes: The gain columns report the median continuation gain from winning, ΔV, in million RWF, by bidder category and type. Median of the discounted continuation-value wedge $\beta\,\Delta V$ divided by the engineering cost estimate, by the bidder's registration category. Sample includes 2,982 bids, with each evaluated as both types at the point estimates.

counterfactuals show how screening affects procurement outcomes through both competition and dynamic incentives.

Each counterfactual replays the same 1,319 auctions among the active firms in the sample. Firms' states—backlog, paid experience, and category—evolve with simulated awards, completion, and promotion. The participation policy is held fixed, but a different set of bidders may participate in each auction. I re-solve each firm's value function under the counterfactual eligibility or promotion policy, evaluate the estimated conditional bid distribution at the simulated states and bidder counts, and adjust bids by the implied change in the continuation value of winning. This approximation allows bids to respond to competition through the estimated bidding function and to future access through continuation values, without jointly re-solving bidding strategies and value functions. As the baseline for comparison, I simulate the status-quo under the existing eligibility and promotion rules. The status quo therefore refers to simulated outcomes under the existing rules, rather than

the observed outcomes. I report average outcomes across 200 replications for each policy in the main comparisons.

The main outcomes of interest are procurement spending, project delays, and the allocation of contracts across firm types. Spending is measured by the change in total spending on lots awarded both in the counterfactual and in the status quo within each replication. Delays are compared on these same projects, using the government's engineering cost estimates (ECEs) as weights, so that delays on larger projects receive greater weight. I also report the share of all projects receiving an award, by number and by ECE. Allocation is measured by the share of total awarded ECE won by high-type firms, 78.2 percent under the status quo. This allocation measure uses all projects awarded under each policy.

For project ℓ, awarded in quarter t to firm i with type θ_i and backlog B_{it}, I construct the predicted unfinished-work path as

$$\varrho_{\ell,u+1} = \varrho_{\ell u}\big[1 - \bar{\xi}(X_{\ell u}, \theta_i)\big], \qquad X_{\ell,u+1} = \big[1 - \bar{\xi}(X_{\ell u}, \theta_i)\big] X_{\ell u}, \tag{49}$$

where u counts quarters since award, $\varrho_{\ell u}$ is the project's unfinished share, and $X_{\ell u}$ is the projected remaining portfolio workload. Initially, $\varrho_{\ell 0} = 1$ and $X_{\ell 0} = B_{it} + z_\ell$. The function $\bar{\xi}(X, \theta_i)$ is the estimated mean portfolio completion share. I measure project delays by

$$\mathcal{T}_\ell^{\text{late}} = \sum_{u=0}^{\infty} \beta^u \varrho_{\ell u}\, \upsilon_{\ell u}. \tag{50}$$

Here, $\upsilon_{\ell u}$ is the fraction of quarter u after award that falls after the project's scheduled completion date (zero for quarters entirely before that date, one for quarters entirely after it, and the remaining fraction for a quarter containing that date). I use the same quarterly discount factor as in estimation, $\beta = 0.95$. Thus, $\mathcal{T}_\ell^{\text{late}}$ is the discounted area under the unfinished-work path beyond the scheduled end; it depends on the winner's type and backlog at award. I average this measure across projects awarded under both policies using their ECEs as weights.

8.1 The cost of screening: loosening eligibility

I relax eligibility in three (increasingly expansive) steps while keeping the award rule unchanged. First, I allow firms one category below to bid on the smaller half of auctions within each category, defined by the median engineering cost estimate. For example, category D firms can bid for the smallest half of category C auctions, in addition to being able to bid for D, E, and F auctions. Second, I expand eligibility to all auctions one category above. Here, category D firms can bid for all category C tenders. Finally, I open all auctions to firms up to two categories below the requirement. Here, category D firms can bid for category C and category B auctions.

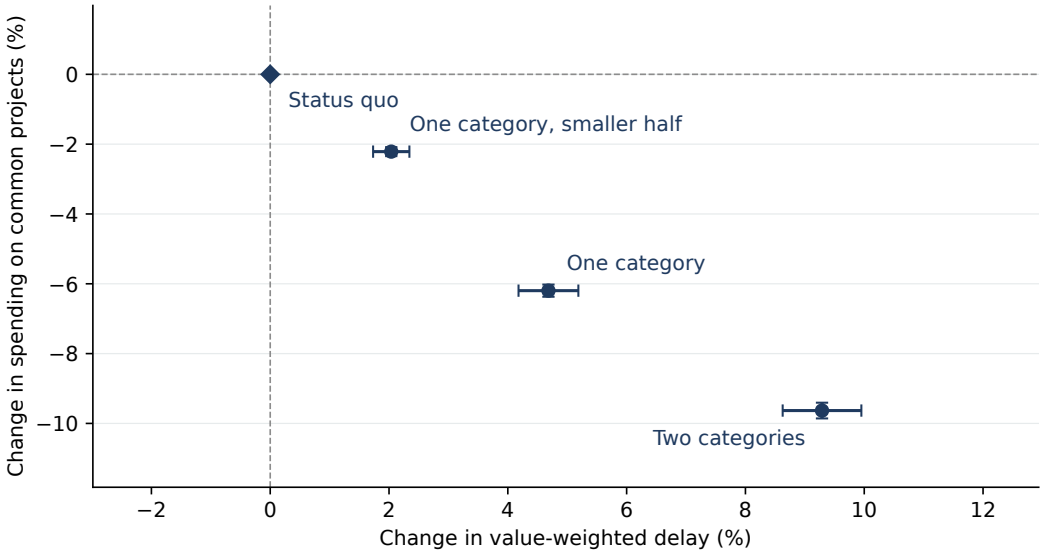
Figure 9 shows that loosening eligibility reduces spending but increases value-weighted delays on projects awarded under both policies. Opening the smaller half of auctions to firms one category below reduces spending by 2.2% and increases delays by 2.0%. As eligibility is extended further,

prices continue to decrease but delays increase. Opening all auctions to firms one and two categories below the requirement reduces spending by 6.2% and 9.6%, but increases value-weighted delays by 4.7% and 9.3%, respectively.

The spending reductions mostly arise from greater competition in high-value projects. In category A and B tenders, the average number of bids per auction rises from 1.28 under the status quo to 1.85 and 2.60 under the one- and two-category eligibility expansions, respectively.

The expansions also change which firms win, which has implications for project completion. The high-type share of awarded value falls by 3.5pp under the first expansion, 7.3pp under the second, and 13.8pp under the third. However, newly eligible winners have substantially less outstanding work. Under the 1-category expansion, newly eligible winners have an average backlog of RWF 0.29 billion, compared with RWF 2.75 billion among winners eligible under the existing rule. Since higher backlog increases completion time, allocating projects to firms with lower backlog partly offsets the increase in delays from allocating projects towards slower firms.

Figure 9: Eligibility expansions: prices and delays



Notes: Auctions in categories E–A; 200 simulations per policy. Spending and delays use only projects awarded under both policies in each replication. Each point shows the average percentage change from the matched simulated status quo. Spending is the sum of winning bids on these projects. Delays measure unfinished work after the scheduled end, weighted by ECE and discounted at $\beta = 0.95$ per quarter. Bars are simulation standard errors across 200 simulations, holding the estimated parameters fixed.

8.2 The value of screening: changing the RPPA promotion rule

Setup. I compare three promotion regimes: the existing RPPA rule, random promotion within each category, and a rule selected to minimize value-weighted delays (the delay-minimal rule). The random promotion rule selects firms at random, holding the per-period category-specific promotion probability fixed to the status quo's. It removes the direct link between completing a contract and ascending up the category ladder and serves as a naïve benchmark to quantify how dynamic

incentives affect firm bidding and project allocation under the status quo. The delay-minimal rule changes the mean category promotion rates and the slope of promotion with respect to paid experience. Both counterfactuals change the dynamic incentives faced by firms, and, through their bids, the allocation of projects across firm types.

For the delay-minimal rule, I search over promotion rules that maintain the estimated logit specification, varying the category intercepts and the coefficient on paid experience relative to the qualification threshold. I do not vary the promotion thresholds themselves. Let $r = (\delta_F, \delta_E, \delta_D, \delta_C, \delta_B, \beta_P^r)$ collect these coefficients, where δ_κ is the change in category κ's intercept and β_P^r is the coefficient on $\log(1 + P/T)$. I search over the family $\mathcal{R}$ defined by $\delta_\kappa \in [-1.5, 1.5]$ and $\beta_P^r \in [0, 3]$. The objective is

$$\min_{r \in \mathcal{R}} \quad \mathbb{E}\left[\frac{\sum_{\ell \in \mathcal{C}(r)} z_\ell \mathcal{T}_\ell^{\text{late}}(r)}{\sum_{\ell \in \mathcal{C}(r)} z_\ell} \right]. \tag{51}$$

Here $\mathcal{C}(r)$ is the set of projects awarded under both rule r and the status quo in a matched simulation. The expectation is over the simulated outcomes under the two policies. The delay measure $\mathcal{T}_\ell^{\text{late}}(r)$ counts discounted unfinished work after the scheduled completion date, as defined above. Thus, the objective averages delays on common awarded projects, weighted by ECE. The common project set can differ across comparisons, so each counterfactual is evaluated against the status quo on its own common set. [20] Appendix E.2.1 repeats the counterfactual exercise treating unawarded projects as permanently unfinished and reports results for random promotion and a rule selected to minimize the delay measure that incorporates non-awards.

I evaluate 1,000 promotion rules, with coefficient combinations chosen to span these ranges. I approximate the expectation using 50 matched replications to compare candidates and use 200 matched replications to report the selected rule's outcomes.

Results: Random promotion. Random promotion reduces firms' future benefit of winning a contract, weakening their incentives to bid aggressively. The mean across replications of the median dynamic markdown among high-type bids falls from 46.4 percent of the bid under the status quo to 2.0 percent under random promotion; for low-type bids, it falls from 6.6 to 0.9 percent.

Table 4 reports the main outcomes. Under random promotion, spending on common awarded projects rises by 8.4 percent and value-weighted delays increase by 0.052 quarters (2.7%). High-type firms receive a smaller share of awarded value and projects, and account for a smaller share of promotions. By the end of the sample, their share of firms in categories A and B also falls (Figure 10). Taken together, these results indicate that conditioning future market access on completed work substantially reduces procurement spending and modestly improves project completion relative

[20] Of the projects awarded under the simulated status quo, 2.8 percent receive no award under random promotion, accounting for 5.6 percent of status-quo awarded ECE. For the delay-minimal rule, the corresponding shares are 8.7 and 14.1 percent. These projects are excluded from the comparisons, as are projects awarded only under the counterfactual.

to random promotion. To understand how much further the government could improve outcomes within the existing category system, I next compare the status quo to the delay-minimal rule.

Table 4: Outcomes under alternative promotion rules

	Random promotion		Delay-minimal rule	
	Status quo	Counterfactual	Status quo	Counterfactual
Projects awarded under both policies				
Spending (status quo = 100)	100	108.43 (0.18)	100	84.31 (0.23)
Value-weighted delay (quarters)	2.191 (0.022)	2.242 (0.020)	2.015 (0.021)	1.963 (0.022)
Common projects	1061.9 (1.1)		998.0 (1.2)	
All projects and firms				
Projects receiving an award (%)	82.8 (0.1)	82.9 (0.1)	82.9 (0.1)	78.6 (0.1)
ECE receiving an award (%)	75.3 (0.4)	74.8 (0.4)	75.3 (0.4)	72.9 (0.4)
High-type share of awarded ECE (%)	78.2 (0.3)	75.9 (0.3)	78.4 (0.3)	77.2 (0.3)
High-type share of projects won (%)	43.1 (0.1)	35.6 (0.1)	42.9 (0.1)	45.4 (0.1)
Firms in A and B	44.9 (0.3)	45.0 (0.3)	44.4 (0.2)	77.6 (0.2)
High-type firms in A and B	28.2 (0.2)	25.3 (0.2)	28.3 (0.2)	43.0 (0.1)
Promotions	111.0 (0.8)	93.3 (0.8)	111.0 (0.8)	218.6 (0.8)
High-type share of promotions (%)	48.3 (0.3)	31.4 (0.4)	48.0 (0.4)	52.7 (0.2)
Replications	200		200	

Notes: Each policy is compared with a simulated status quo using the same random draws. Spending and delays use only projects awarded under both policies in each replication. Spending is normalized to 100 under the matching status quo. Delays measure unfinished work after the scheduled end, weighted by ECE and discounted at $\beta = 0.95$ per quarter from award. Award coverage uses all 1,319 projects and their total ECE. Allocation uses all awarded projects; firm counts refer to the end of the sample. Simulation standard errors across 200 simulations are in parentheses, holding the estimated parameters fixed.

Results: Delay-minimal rule. The RPPA could substantially reduce procurement costs, and reduce delays slightly, by modifying the rule used to promote firms up the category ladder. Relative to the status quo, the delay-minimal rule reduces spending on common awarded projects by 15.7 percent and value-weighted delays by 0.051 quarters (2.6%).

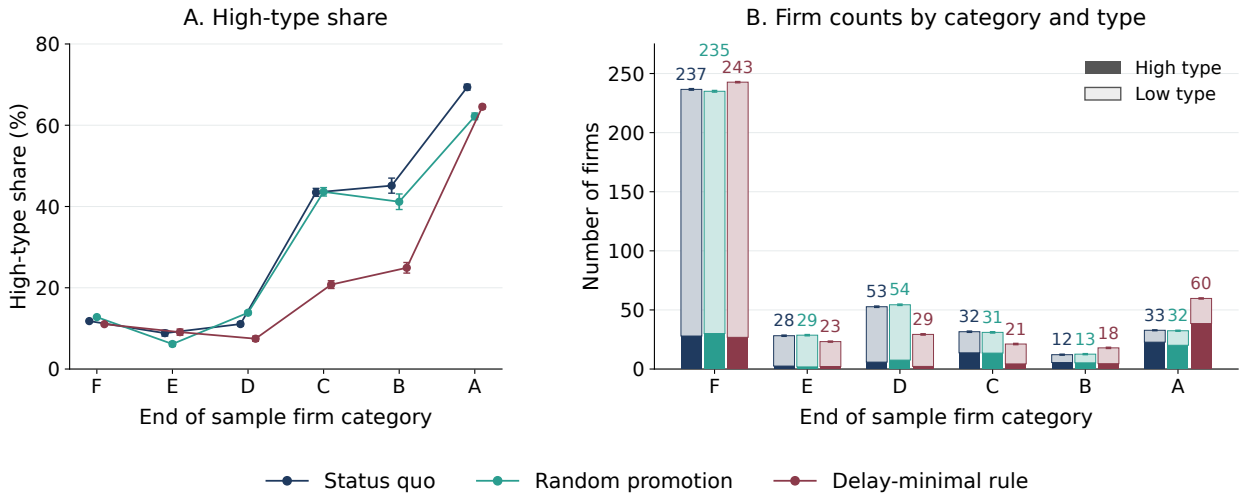
The delay-minimal rule puts more weight on paid experience. Promotion becomes less likely for firms with little paid experience in the two lowest categories (F and E), while firms already in D through B are more likely to advance at a given level of experience. The total number of promotions nearly doubles. Figure 10 shows how this expands the pool of firms eligible for the largest projects, including substantially more high-type firms.

This expansion increases competition for larger projects and pushes prices downward. In category A and B auctions, the average number of qualified bids rises from 1.30 to 2.10. The rule also retains the link between completed work and future access, preserving firms' incentives to bid aggressively to build experience. The median dynamic markdown among high-type bids, averaged across replications, remains substantial at 33.2 percent of the bid, compared with 2.0 percent under random promotion.

Lower backlog among winners explains why delays fall even though the high-type share of all awarded value declines slightly. On common awarded projects, the decomposition shows that

changes in winner type would increase delays. However, winners have less outstanding work, and this reduction in backlog more than offsets the increase.

Figure 10: Composition of the ladder under alternative promotion rules



Notes: Registration categories at the end of the sample, averaged across 200 simulations for each counterfactual. Panel A reports the share of firms that are high-type within each category. In panel B, each bar shows the number of firms, with the solid shaded area representing high-type and the lighter segment low-type firms. Simulation standard errors across 200 simulations, holding the estimated parameters fixed.

9 Conclusion

This paper studies Rwanda's public procurement market for infrastructure contracts, in which the government imposes eligibility rules that condition access to larger contracts on the completion of smaller ones. Project completion records, measured through government payments, reveal persistent differences in the rate at which firms finish awarded work. Completion speed does not enter bid evaluation directly. However, firms that complete work faster accumulate paid experience and move up the category ladder sooner. The category system therefore gives faster firms access to larger projects without requiring the government to observe their type. The model estimates indicate that high-type firms complete their outstanding work about twice as fast and place far more value on winning a tender, all else equal. This larger continuation value produces a larger dynamic bid markdown, which offsets most, but not all, of high-type firms' higher costs. On net, high-type firms bid slightly above otherwise similar low-type firms facing the same tender.

Counterfactual analysis highlights how screening on past performance shapes the price–quality tradeoff. Restricting eligibility reduces competition and raises procurement spending: opening auctions to firms below the required category would reduce spending, but would increase delays. Varying the category promotion rule shows how screening affects prices through firms' incentives to win. Replacing promotion based on completed work with random promotion raises both spending and delays. However, there is still scope to improve the rule. The delay-minimal rule promotes

more firms into the upper categories while retaining the link to completed work, reducing spending substantially and delays slightly on commonly awarded projects.

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A Payment Measurement and Validation

This appendix presents the details behind the construction of the payment-based variables defined in Section 2.5: the component of firm payments attributed to contracts outside the Works portfolio, the contract schedules used for the scheduled-flow measures in the empirical evidence, the persistent firm completion component, and the normalizations that enter the state in the structural model.

A.1 Observed contract portfolios

Let j index public contracts held by firm i. I focus on the set of contracts $\mathcal{W}_i$, which contains signed Umucyo Works contracts. The OCDS data supplies another contract set containing goods, services, and consulting contracts held by the same firms. OCDS contracts are used only to partition firm-level payments; they are not added to the Works backlog studied in the paper.

For each observed contract, let z_j^c denote its VAT-exclusive signed value, and let s_j and e_j denote the quarter containing its recorded start and end dates. I use the signed value to construct the scheduled duration and quarterly flow,

$$d_j^c = e_j - s_j + 1, \qquad r_j = \frac{z_j^c}{d_j^c}.$$

When an end date is missing or precedes the start date, I impute duration using the median valid duration within contract type (sector-category combination). This affects no Works contract and only 64 of the 2,283 OCDS (non-Works) contracts.

A.2 Allocating payments to the Works portfolio

The government withholdings variable in VAT records reports the total payments from public entities, Y_{it}^{tot}, at the firm-quarter level. Because of this, I do not observe contract-level payments. This subsection defines how I attribute payments to Works contracts and portfolios.

I first allocate payments to the firm's observed OCDS non-Works contracts using their values and contract dates. Let Y_{it}^{rem} denote the payment remaining unassigned in this pass, and let t_i^W denote the quarter in which the firm's first observed Works contract begins. Let

$$U_{it} = \min\left\{Y_{it}^{\text{tot}}, \text{median}_{\tau<t_i^W}\, Y_{i\tau}^{\text{rem}}\right\},$$

defining the *unaccounted component* to equal to the "typical" payment the firm received before it held any observed Works contract. For firms with fewer than four such quarters, I set $U_{it} = 0$, attributing their public payments entirely to Works.

The payment that can be attributed to Works is therefore

$$Y_{it}^{\text{tot}} - U_{it}.$$

I attribute this payment to the firm's aggregate Works portfolio rather than to particular

contracts. Let B_{it} be the VAT-exclusive signed-contract value of previously awarded Works that remains unpaid at the beginning of quarter t. For each contract ℓ, let z_ℓ^c denote its signed value net of VAT, and let

$$\Delta W_{it} = \sum_{\ell \in \mathcal{W}_{it}} z_\ell^c$$

be the VAT-exclusive signed-contract value of Works awarded to firm i in that quarter, where $\mathcal{W}_{it}$ is the set of those awards. The value of the firm's open Works portfolio is

$$X_{it} = B_{it} + \Delta W_{it}.$$

I define Works payments as

$$Y_{it}^W = \min\{Y_{it}^{\text{tot}} - U_{it}, X_{it}\}.$$

I attribute payments to Works up to X_{it}, the value of the firm's open portfolio. Payments above this amount remain unallocated; I do not carry them forward to subsequent awards.

In each quarter, Works payments reduce the value of the open portfolio; whatever remains unpaid carries over as the next quarter's backlog:

$$B_{i,t+1} = X_{it} - Y_{it}^W = B_{it} + \Delta W_{it} - Y_{it}^W.$$

Both awarded contract values and payments are measured net of VAT.

Portfolio completion share and paid experience. For firm-quarters with an open portfolio, I measure the quarterly portfolio *completion share* as Works payments received in the quarter divided by the value of the open portfolio:

$$\xi_{it} = \frac{Y_{it}^W}{X_{it}}, \qquad X_{it} > 0.$$

I use this payment-based share as a proxy for the proportion of outstanding work completed during the quarter. It ranges from zero, when none of the portfolio is paid that quarter, to one, when all of it is. I set ξ_{it} to missing when $X_{it} = 0$.

As firms complete work and receive payments, they accumulate *paid experience*, the cumulative value of Works payments they have received. Let W_{i0} denote the cumulative VAT-exclusive signed-contract value of awards before the accounting panel. Then

$$W_{it} = W_{i0} + \sum_{\tau=0}^{t-1} \Delta W_{i\tau}, \qquad P_{i0} = W_{i0} - B_{i0}.$$

At the beginning of quarter t, paid experience is

$$P_{it} = P_{i0} + \sum_{\tau=0}^{t-1} Y_{i\tau}^W = W_{it} - B_{it}.$$

For a firm observed from its first Works award, $P_{i0} = 0$.

A.3 Reduced-form execution measures

All firm-level execution measures in Section 3.2 are built from the quarterly portfolio completion share of Section 2.5,

$$\xi_{it} = \frac{Y_{it}^{W}}{X_{it}},$$

Works payments received in quarter t divided by the portfolio available to receive payment, for firm-quarters with $X_{it} > 0$. Contract schedules enter these measures only as sample filters and workload controls, never as a denominator.

Predetermined completion score. I residualize ξ_{it} on $\log(1+X_{it})$, an indicator for a new award in t, indicators for the number of active Works contracts, and calendar-quarter fixed effects. For each firm I average the residual over its first three quarters with a positive portfolio and standardize the average across firms; firms with fewer than three such quarters are dropped, leaving 319 firms. The score is fixed at the end of a firm's third quarter with positive portfolio.

Explaining variation in completion shares. Table 9 uses ξ_{it} itself on firm-quarters with a positive portfolio and at least 0.005 billion RWF of scheduled Works execution, retaining firms observed in at least three such quarters (1,784 firm-quarters, 233 firms).

Execution quality and tender evaluation. Table 10 uses the predetermined completion score.

A.4 Normalizations entering the state

For the reduced-form regressions the backlog level is divided by the scale of the firm's current category and capped at six; I report both an indicator for any positive backlog and the standardized continuous state. Scheduled backlog, constructed from the administrative contract windows, serves as a comparison measure.

Paid experience enters the promotion problem relative to the requirement stated in the 2014 RPPA categorization manual,

$$\widetilde{P}_{it}(m,\kappa) = \frac{P_{it}}{T_{m,\kappa}},$$

where $T_{m,\kappa}$ is the stated valule requirement for sector m and origin category κ.

A.5 Cross-validating payment-measured Works execution

The measure Y_{it}^{W} assigns the portion of firm i's government payments in quarter t attributable to its Works portfolio observed with the *Umucyo* data. I assess it using independently collected project reports, which record physical completion (as a percent of the project) and the date the report was produced. Project reports are produced by third-party contractors hired to monitor the

project execution and submitted to the relevant procuring entity. I use the date to measure delay in completion relative to scheduled contract start and end dates. I compare portfolio and directly attributable project payments, where I can observe (linking them using the project-specific code).

A.5.1 Payments around independently recorded final reports

Figure 11 plots the share of projects paid at least 75% of their contract value around the recorded final-report quarter. The sample contains 11 contracts for which payments can be attributed directly to the project, with no other contracts immediately before or after. The same projects are observed at every quarter shown. The share paid at least 75% rises from 18% (2 of 11) in the report quarter to 73% (8 of 11) one quarter later and 82% (9 of 11) two quarters later.

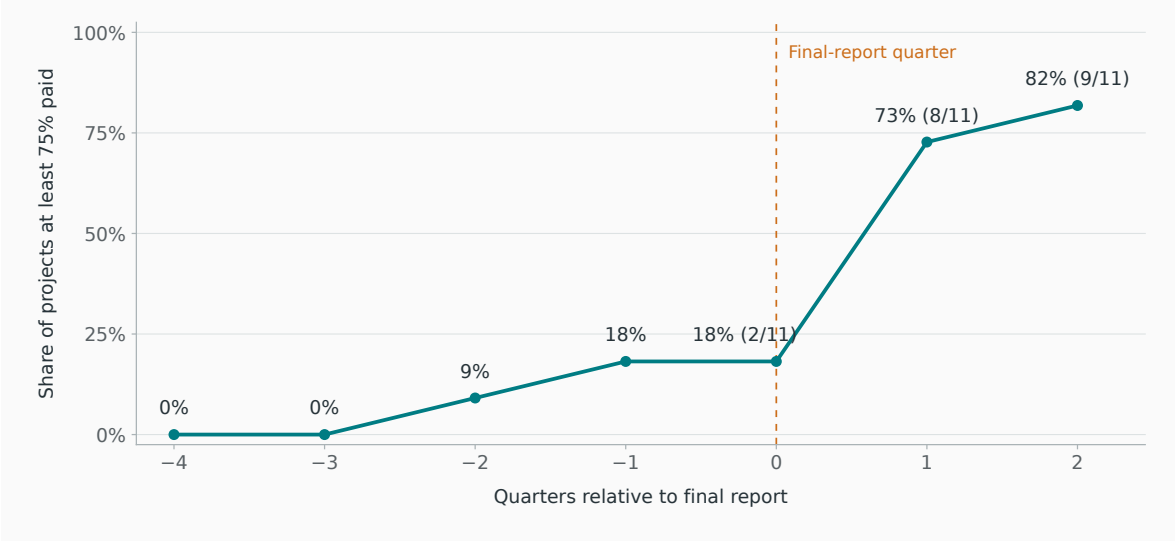


Figure 11: Payments around a recorded final report

Notes: Share of projects with cumulative directly attributable payments of at least 75% of VAT-exclusive contract value. The sample contains the same 11 projects at each quarter, from four quarters before to two quarters after the recorded final report. Each project receives equal weight. Event time is measured in quarters relative to the recorded final-report quarter. One report predates the recorded contract start.

A.5.2 Reported completion and cumulative payments

Table 5 uses the last available report for each matched contract. A 10 percentage point (pp) increase in reported physical completion is associated with a 5.1pp increase in cumulative Y^W, conditional on contract value, scheduled duration, sector, and report year. Contracts coded as delayed are 13.3pp more likely not to have reached 90 percent payment one quarter after the scheduled end, although this estimate is not statistically significant.

A.5.3 Consistency across project and portfolio measures

As a final check, Figure 12 compares the portfolio completion share with an on-time indicator constructed for the subset of contracts whose payment flows can be attributed to individual projects.

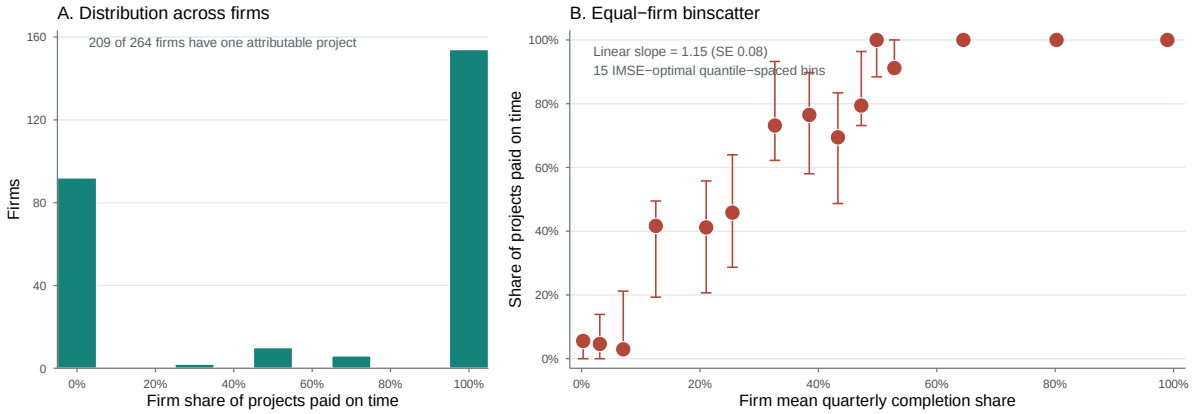
Table 5: Reported project completion and cumulative payments

Payment at report	All reports		Non-final only				
Payment after signed end				Unpaid share		Not 90% paid	
Specification	Raw	Controls	Raw		Controls	Raw	Controls
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Reported completion share	0.641***	0.510**	1.04***				
	(0.157)	(0.221)	(0.152)				
Report-coded delay				-0.022	0.016	0.139	0.133
				(0.070)	(0.071)	(0.098)	(0.095)
Mean dependent variable	0.551	0.551	0.509	0.397	0.397	0.702	0.702
Contract value and duration controls	No	Yes	No	No	Yes	No	Yes
SE clustered by	Firm	Firm	Firm	Firm	Firm	Firm	Firm
Sector fixed effects		✓			✓		✓
Report-year fixed effects		✓					
Start-year fixed effects					✓		✓
Observations	108	108	26	121	121	121	121
Firms	62	62	20	65	65	65	65

Notes: Each contract uses its last recorded report, which may be Final or non-Final (i.e., progress report). Payment at report is cumulative Y^W divided by contract value. End-point outcomes are measured one quarter after the signed contract end. Controlled specifications include contract value, scheduled duration, sector fixed effects, and report- or start-year fixed effects. Standard errors are clustered by firm.

Across 264 firms, the correlation between mean portfolio completion share and the share of projects paid at least 90 percent by scheduled end plus one quarter is 0.675. The correlation is 0.651 among the set of 55 firms with at least two attributable projects. Firms with higher portfolio completion shares also have a larger share of projects paid on time.

Figure 12: Project-level on-time payment and portfolio completion shares



Notes: The sample contains 264 firms, 332 attributable projects, and 2,427 positive-backlog firm-quarters. A project is on time if cumulative Y^W reaches at least 90 percent of VAT-exclusive contract value by scheduled end plus one quarter. The portfolio completion share is the firm's mean quarterly completion share across positive-backlog quarters. Each firm receives equal weight. Of the 264 firms, 209 have only one attributable project, so their on-time payment share is either zero or one (Panel A). Panel B reports the average on-time payment share among firms in each bin. Panel B reports the IMSE-optimal number of quantile-spaced bins (15) in the firm mean portfolio completion share using (Cattaneo et al., 2024) in R; whiskers are 95% CIs.

The validation exercises support using payments as a proxy for execution. Payments rise around independently reported completion, reported physical progress predicts cumulative payments, and firms' portfolio completion shares are positively correlated with their project-level on-time payment shares.

B Supporting Empirical Evidence

B.1 Firm categorization requirements

The RPPA category assigned to a firm reflects four documented forms of capacity: average construction turnover over the previous five years, executed-contract references, owned equipment, and permanent senior staff. Table 6 reports the turnover and past executed-tender requirements for the three sectors studied in the paper: Buildings, Roads and Bridges, and Drinking Water Supply. These requirements applied to firms requesting a category during the sample period.

Table 6: Partial requirements for firm categorization

Category	Tender value	Five-year average turnover	Executed tenders: # and value of each
A	$>$ 2 bn	$>$ 1.5 bn	2; each $>$ 1.2 bn
B	1.5–2 bn	1.0–1.5 bn*	2; each 0.9–1.2 bn*
C	0.8–1.5 bn	0.6–1.0 bn*	2; each 0.48–0.90 bn*
D	0.3–0.8 bn	0.25–0.60 bn	1; 0.18–0.48 bn
E	0.1–0.3 bn	0.075–0.25 bn	1; 0.06–0.18 bn
F	$<$ 0.1 bn	None	None

Notes: Values are in billions of RWF. Stars identify the exceptions summarized here. For Drinking Water Supply, the turnover requirement is 1.1–1.5 billion RWF in category B and 0.6–1.1 billion RWF in category C; the category-C executed-tender requirement is 0.50–0.90 billion RWF. Executed tenders must fall within the previous five years. Roads and Bridges divides categories A and B into A1/A2 and B1/B2. A1 and B1 require asphalt-road project references, whereas A2 and B2 require road-construction project references; their numerical thresholds are the same. The category-B reference cell is marked to flag this distinction. The manual also specifies asphalt-specific equipment and staff experience for A1 and B1, and owned equipment and permanent senior staff in all sectors. Source: Rwanda Public Procurement Authority (2014).

B.2 Category eligibility binds in the bidding data

Table 7 shows results whether the tender eligibility requirements are enforced. The sample includes all National and International Competitive Bidding, NCB and ICB respectively, bids in the three main sectors, Buildings, Roads and Bridges, and Drinking Water Supply, before October 2023. In the combined sample, 99.5 percent of bids come from firms in the tender category or no more than two categories above it, and the same is true of winning bids.

Table 7: Category eligibility among observed bidders

Sector	Bids	Lots	Firms	Eligible bids	Same cat.	One above	Two above
All main sectors	7,034	1,704	766	99.5	67.0	20.1	12.4
Buildings	5,190	1,172	696	99.6	71.1	16.9	11.5
Roads and Bridges	850	287	152	98.7	53.3	27.3	18.1
Drinking Water Supply	994	245	184	99.6	56.8	30.8	12.0

Notes: NCB and ICB bids in the three main Works sectors before October 2023, restricting to observations with classified tender and firm categories. "Eligible bids" is the share of bids in which the firm is in the tender category or one or two categories above it. The final three columns decompose the bidder–tender category relationship.

B.3 Paid experience and approval of category-upgrade requests

Table 8 compares firms requesting the same sector-specific category transition in the same quarter. Applicants with more cumulative paid Works experience are more likely to be approved. Meeting the stated value requirement for the requested category is associated with a 10.3 percentage point higher approval probability, relative to an 88.5 percent mean.

Table 8: Paid experience and approval of category-upgrade requests

Outcome	Category-upgrade request approved			
	(1)	(2)	(3)	(4)
Log(1 + paid experience / target requirement)	0.056**	0.114**		
	(0.028)	(0.054)		
Paid experience meets target requirement			0.103***	0.151*
			(0.036)	(0.084)
Firm experience controls	No	Yes	No	Yes
Request-quarter × sector × origin × target FE	Yes	Yes	Yes	Yes
SE clustered by	Firm	Firm	Firm	Firm
Observations	338	338	338	338
Mean dependent variable	0.885	0.885	0.885	0.885
Request events	201	201	201	201
Firms	164	164	164	164

Notes: The unit of observation is a sector-specific category-upgrade request. Paid experience is cumulative Works payments, P_{it}, measured by the quarter before the request. The target requirement is the stated value requirement for the requested sector and category. Firm experience controls include firm age, prior observed Works contracts, and prior category requests. Standard errors are clustered by firm.

B.4 Sources of variation in portfolio completion shares

What explains variation in portfolio completion shares? Table 9 reports the Adjusted R^2 of an increasingly rich set of covariates. The regression specification is

$$\xi_{it} = \delta_t + \alpha_i + (\mathbf{x}_{it}^W)'\boldsymbol{\beta}_W + (\mathbf{x}_{it}^C)'\boldsymbol{\beta}_C + (\mathbf{x}_{it}^P)'\boldsymbol{\beta}_P + (\mathbf{x}_{it}^F)'\boldsymbol{\beta}_F + \varepsilon_{it}, \tag{52}$$

where $\mathbf{x}_{it}^W$ contains portfolio workload controls, $\mathbf{x}_{it}^C$ procuring-entity composition controls, $\mathbf{x}_{it}^P$ prior procuring-entity payment controls, and $\mathbf{x}_{it}^F$ firm size and experience controls. The terms δ_t and α_i are quarter and firm fixed effects. I estimate these regressions by OLS, adding the four control groups in this order and including firm fixed effects only in the final specification.

I begin with quarter fixed effects. I add "Portfolio workload" controls, which include log outstanding work after new awards, an indicator for a new award, and indicators for the number of active contracts. This substantially increases the Adjusted R^2 to 0.145. I then add Procuring entity (PE) composition covariates, which include the share of the firm's portfolio value contracted with a new PE, the share of contracts located in the firm's own district, indicators for the number of procuring entities that the firm is contracting with, and a HHI index of value across procuring entities. These covariates aim to summarize the newness of the firm's relationships with the PEs

and how concentrated the firm's contracts are. Next, I add prior PE payment controls, which include the value-weighted mean of each PE's earlier payment shares for contracts between the PE and *other* firms. These are payer-side controls that aim to control for PE payment delays to firms. Despite the richness in these controls, the adjusted R^2 only increases to 0.149. Adding firm-level covariates, including category, prior-year VAT sales, paid experience, prior Works contracts, and number of quarters in the market, increases the Adjusted R^2 modestly to 0.193. Finally, adding firm fixed effects more than doubles the adjusted R^2 to 0.431, providing suggestive evidence that the completion share contains an important unobserved firm-specific component.

Table 9: Explaining variation in portfolio completion shares

Specification	Adjusted R^2
Quarter fixed effects	0.034
+ Portfolio workload	0.145
+ PE composition	0.148
+ Prior PE payment	0.149
+ Firm size and experience	0.193
+ Firm fixed effects	0.431
Observations	1,784
Firms	233

Notes: The outcome is the quarterly portfolio completion share $\xi_{it} = Y_{it}^W / X_{it}$. The sample contains firm-quarters with a positive Works portfolio and at least 5 million RWF of scheduled Works execution, retaining firms observed in at least three such quarters.

B.5 Payment-measured execution quality and tender evaluation

Table 10 tests whether the portfolio completion score predicts disqualification process and bid award. I use bid-level Linear Probability Model regressions to test whether the predetermined portfolio completion score predicts disqualification for any reason, disqualification for technical reasons, and contract award (lowest bid among qualified). For award, I also include bid-rank Fixed Effects. The coefficient for disqualification for any reason is negative and marginally statistically significant, but all other regressions show no evidence that the completion score (the proxy for firm quality) predicts disqualification or bid award. Conditional on the price ranking that determines award, firms with higher completion shares are no more likely to win.

Table 10: Payment-measured execution quality and tender evaluation

	Any disq. (1)	Technical disq. (2)	Award (3)	Award (4)
Predetermined completion score	-0.031*	-0.014	-0.027	-0.001
	(0.017)	(0.012)	(0.030)	(0.006)
Bid-category controls	Yes	Yes	Yes	Yes
Current scheduled backlog	Yes	Yes	Yes	Yes
Tender-lot FE	Yes	Yes	Yes	Yes
Bid-rank FE	No	No	No	Yes
Sample	Entrants	Entrants	Qualified	Qualified
Clustered SEs	Firm	Firm	Firm	Firm
N	3,216	3,216	1,742	1,742
N firms	183	183	161	161

Notes: Bid-level LPMs. The quality proxy is the standardized predetermined portfolio completion score of Appendix A.3, fixed at the end of a firm's third positive-portfolio quarter. The regression sample includes bids submitted at or after the quarter in which the bidder's score is calculated. Columns 1 and 2 use all bids, while columns 3 and 4 use bids were not disqualified. Relative to column 3, column 4 includes bid rank Fixed Effects. All specifications absorb tender-lot fixed effects and control for bidder category, the bidder–tender category gap, and current scheduled backlog. Standard errors are clustered by firm.

C Estimation Details

C.1 Detailed State Accounting

This appendix describes the state panel used by the model. Let ΔW_{it} be the total VAT-exclusive signed value of Works contracts awarded to firm i in quarter t, so $\Delta W_{it} = \sum_{\ell \in \mathcal{W}_{it}} z_\ell^c$, and let $\widehat{Y}_{it}^W$ be RRA payments allocated to those Works contracts. Starting from the observed stock B_{it} at the beginning of the quarter, I construct

$$\begin{aligned} X_{it} &= B_{it} + \Delta W_{it}, \\ Y_{it}^W &= \min\{\widehat{Y}_{it}^W, X_{it}\}, \\ B_{i,t+1} &= X_{it} - Y_{it}^W, \\ W_{i,t+1} &= W_{it} + \Delta W_{it}, \\ P_{i,t+1} &= W_{i,t+1} - B_{i,t+1}. \end{aligned}$$

The initial cumulative-award stock is the VAT-exclusive signed value of observed awards before the estimation window, subject to $W_{i0} \geq B_{i0}$. I set $P_{i0} = W_{i0} - B_{i0}$; hence the recursion gives

$$P_{it} = P_{i0} + \sum_{\tau=0}^{t-1} Y_{i\tau}^W = W_{it} - B_{it}.$$

Thereafter the recursion uses awards and payments only. I verify the recursion against the independently constructed backlog panel in every overlapping firm-quarter.

Constructing the payments this way has two important timing properties. First, an award enters neither B_{it} nor W_{it} before its award quarter. Second, its full VAT-exclusive signed value enters X_{it} before the quarter's completion-share draw. Thus, realized commitments and payments use the same VAT-exclusive signed contract value. The continuation value's win–lose calculation is different: because z_ℓ^c is not observed when bids are submitted, it uses the engineering estimate z_ℓ as a predetermined proxy for the prospective award increment, similar to (De Silva et al., 2023). Under this approximation, the continuation gain is fixed with respect to the bid.

C.2 Execution-type likelihood

The three components of (31) are logits in the standardized log outstanding work,

$$x_{it}^\xi = \frac{\log X_{it} - \overline{\log X}}{\mathrm{sd}(\log X)},$$

where the mean and standard deviation are taken over the risk-set quarters, those with $X_{it} > 0$. Execution type enters each component through the single common loading λ^ξ, as in (32). Standardizing $\log X_{it}$ absorbs the mechanical dependence of the completed share on the size of the stock, so λ^ξ is not identified from firm scale. Restricting the three type effects to a single λ^ξ with

this sign pattern is what delivers the stochastic ordering (9), and fixes which mixture component is denoted H.

For each observation in the risk set, i.e., has outstanding work $X_{it} > 5$mill. RWF, the likelihood uses the quarterly completion share $\xi_{it} \in [0,1]$ from (7). Let $f_{\mathrm{B}}(\cdot; a, b)$ denote the Beta density. Under (31) and (32), the execution component of the firm likelihood is

$$\mathcal{L}_i^{\xi}(\theta) = \prod_{t: X_{it}>0} \begin{cases} p_{0\theta}(X_{it}), & \xi_{it} = 0, \\ [1 - p_{0\theta}(X_{it})] p_{1\theta}(X_{it}), & \xi_{it} = 1, \\ [1 - p_{0\theta}(X_{it})][1 - p_{1\theta}(X_{it})] f_{\mathrm{B}}\left(\xi_{it}; \mu_\theta^{\xi}(X_{it})\phi, [1 - \mu_\theta^{\xi}(X_{it})]\phi\right), & 0 < \xi_{it} < 1. \end{cases}$$

The common parameter $\lambda^{\xi} > 0$ implies

$$p_{0H}(X) < p_{0L}(X), \qquad p_{1H}(X) > p_{1L}(X), \qquad \mu_H^{\xi}(X) > \mu_L^{\xi}(X).$$

That is, conditional on the same level of outstanding work, high-type firms have a lower probability of completing 0% of work, a higher probability of completing 100% of work, and, conditional on neither of the above, a higher average completion share.

The Beta precision ϕ is common across types. Repeated zero payments, partial payments, and full payment, conditional on the total remaining portfolio, therefore all contribute to type classification.

C.3 Sector-specific categories and sector entry

Firm-level total backlog and payment are shared across sectors, while registration is sector-specific. So, experience gained in Buildings can contribute to a higher probability of upgrading in Roads & Bridges. For every active sector m, the RPPA category upgrade policy is

$$\frac{P_{it}}{T_{m,\kappa_{im,t}}}$$

and updates the firm's category in that sector, $\kappa_{im,t}$ using the sector-specific payment threshold $T_{m,\kappa_{im,t}}$. While the thresholds are allowed to be sector-specific, in practice they are almost all equal across sectors (see Table 6).

C.4 Exogenous policy estimates

This appendix collects the policies the model takes as given rather than recovers from optimizing behavior: the RPPA category transition kernel, the industry entry and exit hazards, and the aggregate state and its law of motion. Each is estimated once, on its own sample, and then held fixed for the remainder of estimation. Bootstrap intervals do not account for the uncertainty in the estimation of these policies. The estimated primitives — the completion process, the participation policy, and the cost equation — are reported in Appendix E.

C.4.1 RPPA category transitions

The promotion hazard (29) and destination rule (30) are estimated on observations at the firm–sector–quarter–level, excluding category A, which cannot be promoted. The sample used for estimation contains 5,081 firm–sector–quarters covering 357 firms.

Table 11 reports the fitted promotion hazard by origin category and paid-experience ratio. The hazard rises steeply in $\widetilde{P}$ throughout, and it does so smoothly. Crossing the stated requirement raises the promotion probability substantially, but not to probability 1. This is consistent with the legal criteria also requiring contract counts, equipment, and professional staff that I do not observe.

Table 11: Fitted quarterly promotion hazard, by origin category and paid experience

	Paid experience relative to requirement, $\widetilde{P}$						
Origin	0	$(0, 0.5]$	$(0.5, 1]$	$(1, 2]$	$(2, 4]$	$(4, 8]$	> 8
F	0.005	0.006	0.009	0.015	0.029	0.062	0.110
E	0.006	0.007	0.011	0.018	0.034	0.072	0.127
D	0.012	0.015	0.023	0.038	0.071	0.143	0.238
C	0.019	0.023	0.037	0.059	0.108	0.208	0.330
B	0.025	0.030	0.047	0.076	0.136	0.256	0.392

Notes: Quarterly probability of moving to any higher category. Entries are the sector-pooled policy, which is used in the value computation. Sector-specific hazards are also estimated and are used where a sector-specific transition is required. Category A is omitted because it cannot be promoted.

Conditional on promotion, Table 12 reports where firms go. Firms primarily move up to the adjacent category from E and D, while promotions out of F and C are more dispersed. Upgrades from categories E and D are restricted to at most two ranks. Moves from F are unrestricted.

Table 12: Destination shares conditional on promotion

	Destination category				
Origin	E	D	C	B	A
F	0.599	0.237	0.101	0.055	0.009
E		0.928	0.072		
D			0.892	0.108	
C				0.589	0.411
B					1.000

Notes: Shares average the sector-pooled destination distribution across paid-experience bins. Rows sum to one. Blank cells are destinations the policy does not assign positive probability.

C.4.2 Industry entry and exit

Table 13 reports the entry and exit fitted hazards. Exit falls with backlog and, more steeply, with paid experience. Entry is a function of the aggregate state (new firms have no individual state).

These hazards are exogenous policy functions used in the value computation. The value function in (22) is conditional on the firm remaining in the market; the continuation expectation incorporates

Table 13: Industry entry and exit hazards

	Entry	Exit
Constant	−2.452	−1.898
Category E		−0.636
Category D		−0.703
Category C		−0.724
Category B		−0.469
Category A		−0.102
$\log(1 + B_{it})$		−0.371
$\log(1 + P_{it})$		−0.760
Aggregate state, A	−0.064	−0.096
Aggregate state, B	−0.054	−0.019
Aggregate state, C	0.002	0.070
Aggregate state, D	0.376	0.073
Aggregate state, E	−0.225	−0.198
Firm-quarters	7,916	4,726
Event rate	0.081	0.100
Log loss	0.274	0.310
Brier score	0.073	0.087
Holdout firm-quarters	579	1,083
Holdout log loss	0.444	0.358
Holdout Brier score	0.130	0.104

Notes: Table reports logit coefficients for the entry and exit hazards as defined above. Category F is the omitted category in the exit model; the entry model conditions on the aggregate state alone, since the risk set is defined before a firm's first Works bid. Aggregate-state components are the standardized four-quarter qualified-bidder moments of (4). Values are measured in billions of Rwandan francs.

exit before the next quarter's auctions. Appendix D.3 relates this convention to the values stored before the exit draw in the numerical implementation. The firm's final bid date is used to measure its exit. The sample for structural estimation ends in 2023Q3, but the full sample ends in 2025Q3, so I use the quarters after the structural estimation to determine whether the firm bids again. A firm who bid twice by 2023Q3 and at least once between 2023Q3 and 2025Q3 is not coded as having exited.

C.4.3 The aggregate state and its law of motion

The aggregate state summarizes the distribution of potential competitors by a small number of moments that firms use to forecast future auction competition. I use the average number of qualified bidders in each tender category, for categories E–A (those that firms can upgrade to).

Let $\mathcal{A}_\tau(k)$ be the set of Works lots in category $k \in \{A, B, C, D, E\}$ auctioned in quarter τ, and let

$$N_\ell^q = \#\{j : \ j \text{ submits a bid on } \ell \text{ and is not disqualified}\}.$$

At the beginning of quarter t, define

$$\bar{n}_t^k = \frac{\displaystyle\sum_{\tau=t-4}^{t-1} \sum_{\ell \in \mathcal{A}_\tau(k)} N_\ell^q}{\displaystyle\sum_{\tau=t-4}^{t-1} |\mathcal{A}_\tau(k)|}, \qquad k \in \{A, B, C, D, E\},$$

and collect these five numbers in $\bar{s}_t = (\bar{n}_t^A, \ldots, \bar{n}_t^E)$. I construct the aggregate state using quarters $t-4$ through $t-1$, so it is predetermined when firms make their current participation and bidding decisions. If there are no category-k auctions during this window, I use the mean over all earlier auctions in that category. I set the component to zero before the first category-k auction.

The participation, bid, entry, and exit specifications use standardized $\log(1 + \bar{n}_t^k)$. For the conditional-choice-probability step, I divide each log component into low, medium, and high values using its time-series terciles. I use only the 22 distinct combinations that ever occur during the 24 sample quarters, instead of the possible $3^5 = 243$ possible combinations. I fit a multinomial logit for the probability that each category bin moves to low, medium, or high next quarter, as a function of all 5 current bins. The joint transition is the (normalized) product of the five component probabilities. Figure 13 shows how the five components evolve over the estimation window. The series are persistent within category but not across categories, illustrating varying degrees of competition across categories.

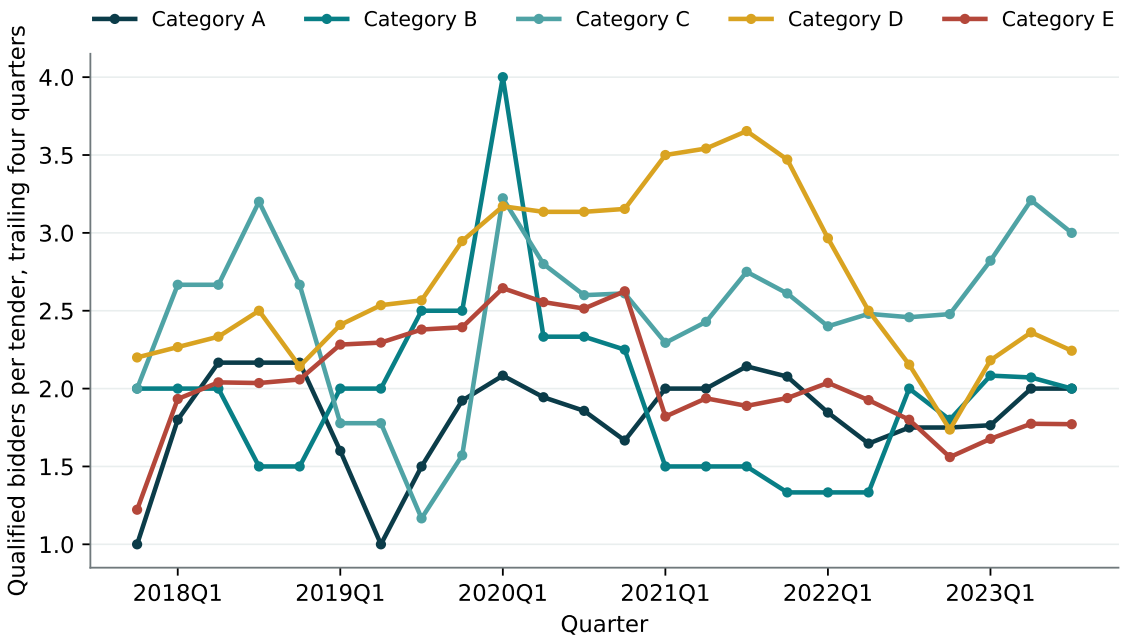


Figure 13: Evolution of the aggregate state

Notes: Each series is a component of $\bar{s}_t$: the mean number of qualified bidders per tender in that category over quarters $t-4,\ldots,t-1$. Category B is the category with least number of auctions, averaging 3.9 tender-lots per four-quarter window compared with 39.5 for category E, so its component is the most volatile.

C.5 Type posterior and bid policy

C.5.1 Observed-data likelihood

For firm i, let $\mathcal{D}_i^{\xi}$, $\mathcal{D}_i^{a}$, and $\mathcal{D}_i^{b}$ denote its execution, participation, and bid histories. Let $\mathcal{C}_i$ collect the initial measured state and the opportunity and tender characteristics treated as conditioning variables. The state s_{it} appearing in each contribution is predetermined relative to that outcome: it is constructed recursively from the initial state and outcomes observed before the current draw.

Conditional on permanent type, the chain rule gives

$$\begin{aligned} &p\Big(\mathcal{D}_i^{\xi}, \mathcal{D}_i^{a}, \mathcal{D}_i^{b} \mid \theta_i = \theta, \mathcal{C}_i\Big) \\ &\quad = p\Big(\mathcal{D}_i^{\xi} \mid \theta_i = \theta, \mathcal{C}_i\Big)\, p\Big(\mathcal{D}_i^{a} \mid \mathcal{D}_i^{\xi}, \theta_i = \theta, \mathcal{C}_i\Big) \\ &\qquad \times p\Big(\mathcal{D}_i^{b} \mid \mathcal{D}_i^{a}, \mathcal{D}_i^{\xi}, \theta_i = \theta, \mathcal{C}_i\Big)\,. \end{aligned} \tag{53}$$

Conditional on the firm's current state, auction characteristics, and permanent type, I assume that transitory completion shocks are independent of participation and bid shocks. Transitory shocks are also independent across quarters and auctions. A bid is observed only when the firm

submits a qualified bid, i.e., $a_{i\ell t} = 1$. Disqualified bids are excluded. It follows that

$$\begin{aligned} & p\Big(\mathcal{D}_i^{\xi}, \mathcal{D}_i^{a}, \mathcal{D}_i^{b} \mid \theta_i = \theta, \mathcal{C}_i\Big) \\ &= \prod_{t \in \mathcal{D}_i^{\xi}} f_{\theta}^{\xi}(\xi_{it} \mid X_{it}) \\ &\quad \times \prod_{(\ell,t) \in \mathcal{D}_i^{a}} \left[p_{\theta,i\ell t}^{a}\right]^{a_{i\ell t}} \left[1 - p_{\theta,i\ell t}^{a}\right]^{1 - a_{i\ell t}} \\ &\quad \times \prod_{(\ell,t) \in \mathcal{D}_i^{b}} f_{\theta}^{b}\Big(\widetilde{b}_{i\ell t} \mid a_{i\ell t} = 1, \chi_{\ell}, s_{it}, \bar{s}_t\Big) \\ &= \mathcal{L}_i^{\xi}(\theta) \mathcal{L}_i^{a}(\theta) \mathcal{L}_i^{b}(\theta), \end{aligned} \tag{54}$$

where

$$p_{\theta,i\ell t}^{a} = a_{\theta}(\chi_{\ell}, s_{it}, \bar{s}_t).$$

Equation (53) follows from the chain rule. The product representation in (54) follows from the maintained Markov and conditional-independence restrictions.

The baseline specification uses a constant mixing share,

$$\mathbb{P}(\theta_i = \theta \mid \mathcal{C}_i) = \omega_{\theta}.$$

Summing over permanent types therefore gives

$$\mathcal{L}_i = \sum_{\theta \in \{L,H\}} \omega_{\theta} \mathcal{L}_i^{\xi}(\theta) \mathcal{L}_i^{a}(\theta) \mathcal{L}_i^{b}(\theta). \tag{55}$$

Backlog evolves mechanically from the initial state, new awards, and observed completion. The exogenous policies – category promotion, entry, exit, tender-arrival, and aggregate-state – are estimated separately.

C.5.2 Econometrician's type posterior

The participation outcome is = 1 only when the firm's submitted bid is not disqualified. The econometrician's posterior is

$$w_i = \mathbb{P}(\theta_i = H \mid \mathcal{D}_i^{\xi}, \mathcal{D}_i^{b}, \mathcal{D}_i^{a}). \tag{56}$$

D Auction Inversion and Value Computation

D.1 Category distribution beliefs and the bid hazard

I parametrize normalized bids to follow a type-specific conditional Gamma function whose conditional mean depends on normalized backlog, $\log(1 + \tilde{P})$, the aggregate state, project scale, bidder count, sector, quarter, and firm-category–tender-category cells. Firm i knows its own bid function (because

it knows θ_i), but not rivals' bid functions. Let $f_{\theta,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t)$ denote the type-specific bid density and let $\bar{F}_{\theta,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t) = 1 - F_{\theta,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t)$ denote its survival function.

For lot ℓ, set $m = m_\ell$, so rival j's eligibility category is $\kappa_{jm,t}$. Let $\omega_{H,\kappa_{jm,t}}$ be the estimated share of type-H firms in category κ. These six category shares are common prior beliefs and are constructed from the estimated posterior masses in the relevant eligibility categories. Once rival j is observed in the entrant set, the prior is reweighted by the type-specific participation policy as in (17). Write the resulting probability as $\tilde{\omega}_{j\ell t}$. For rival j, define the perceived density and survival function

$$\begin{aligned} f_j^{\text{mix}}(b \mid \chi_\ell, s_{jt}, \bar{s}_t) &= \tilde{\omega}_{j\ell t} f_{H,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t) \\ &\quad + [1 - \tilde{\omega}_{j\ell t}] f_{L,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t), \\ \bar{F}_j^{\text{mix}}(b \mid \chi_\ell, s_{jt}, \bar{s}_t) &= \tilde{\omega}_{j\ell t} \bar{F}_{H,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t) \\ &\quad + [1 - \tilde{\omega}_{j\ell t}] \bar{F}_{L,j}(b \mid \chi_\ell, s_{jt}, \bar{s}_t). \end{aligned} \tag{57}$$

The perceived contribution of rival j to the procurement hazard is the hazard of this mixture. Thus, bidder i's hazard is

$$H_i(b \mid \mathcal{J}_{\ell t}) = \sum_{j \in \mathcal{J}_{\ell t} \setminus \{i\}} \frac{f_j^{\text{mix}}(b \mid \chi_\ell, s_{jt}, \bar{s}_t)}{\bar{F}_j^{\text{mix}}(b \mid \chi_\ell, s_{jt}, \bar{s}_t)}. \tag{58}$$

Writing $h_{\theta,j} = f_{\theta,j}/\bar{F}_{\theta,j}$ for the type-specific hazards,

$$\frac{f_j^{\text{mix}}(b)}{\bar{F}_j^{\text{mix}}(b)} = \tilde{\omega}_{j\ell t} \frac{\bar{F}_{H,j}(b)}{\bar{F}_j^{\text{mix}}(b)} h_{H,j}(b) + \big(1 - \tilde{\omega}_{j\ell t}\big) \frac{\bar{F}_{L,j}(b)}{\bar{F}_j^{\text{mix}}(b)} h_{L,j}(b), \tag{59}$$

a survival-function-weighted average of the type hazards whose weights depend on the bid.

D.2 Derivation of the bid first-order condition

From the value function to the bid objective. Fix a project ℓ in which firm i participates. The bid is chosen after the firm observes its cost, the entrant set, and its rivals' public states. Hold the firm's actions and outcomes in its other quarter-t tenders fixed, and split the expectation in (22) by the outcome of the focal auction: with probability $G_i(b \mid \mathcal{J}_{\ell t})$ the firm wins, collects $b - c_{i\ell t}$, and continues from $s_{i,t+1}^{\text{win}}$; otherwise it continues from $s_{i,t+1}^{\text{lose}}$. The terms involving b are

$$\begin{aligned} &G_i(b \mid \mathcal{J}_{\ell t}) \Big[b - c_{i\ell t} + \beta\, \mathbb{E}\, V_{\theta_i}\big(s_{i,t+1}^{\text{win}}, \bar{s}_{t+1}\big) \Big] + \big[1 - G_i(b \mid \mathcal{J}_{\ell t})\big]\, \beta\, \mathbb{E}\, V_{\theta_i}\big(s_{i,t+1}^{\text{lose}}, \bar{s}_{t+1}\big) \\ &\quad = G_i(b \mid \mathcal{J}_{\ell t}) \big[b - c_{i\ell t} + D_{i\ell t} \big] + \beta\, \mathbb{E}\, V_{\theta_i}\big(s_{i,t+1}^{\text{lose}}, \bar{s}_{t+1}\big), \end{aligned}$$

where the equality adds and subtracts the losing continuation value and applies the definition of $D_{i\ell t}$ in (23). The losing continuation does not depend on b, and flow profits from the firm's other quarter-t tenders enter additively. Both continuation expectations include exit before the next quarter's auctions, with zero value upon exit. Dropping terms that do not depend on b yields the

objective (24).

First- and second-order conditions. Suppress indices and write the expected payoff from bid b as

$$u(b) = G(b)(b - c + D),$$

where $G'(b) < 0$ in a low-price auction. The engineering cost estimate increment and completion transition process are both fixed before bidding, so the continuation gain satisfies $\partial D/\partial b = 0$. The first-order condition is

$$G(b) + G'(b)(b - c + D) = 0.$$

Therefore,

$$b - c + D = -\frac{G(b)}{G'(b)} = \frac{1}{-G'(b)/G(b)} = \frac{1}{H(b)},$$

which yields (25). The second-order condition is

$$2G'(b) + G''(b)(b - c + D) < 0.$$

The empirical support restricts attention to bids for which the first-order inversion is well defined and the estimated rival hazard is positive.

D.3 Value computation

The value computation follows the procedure of (De Silva et al., 2023), adapted for discrete time (their model is in continuous time). Let M_θ^0 denote the "no-award" transition matrix for type θ under the estimated policies. It integrates over completion, payments, paid experience, category promotion, and the aggregate-state law of motion, followed by exit before the next quarter's auctions, while setting new awards to zero. The zero-valued exit state is omitted, so the row sums give the probability of remaining in the market next quarter. Let $\mathbf{\Pi}_\theta^A$ stack the expected quarterly auction surplus in (45), conditional on remaining in the market at the start of the current quarter. The expectation is over participation, project arrivals and characteristics, the firm's own bid, and the realized rival set under the estimated policies.

This expectation is estimated from the observed bids. The per-bid ratio, $\widehat{G}/\widehat{H}$, is regressed on project characteristics, the firm's state, the aggregate state, and type, and the fitted conditional mean, multiplied by the participation probability and the arrival intensities, is evaluated on the grid. By the first-order condition above, $G(b)/H(b) = G(b)(b - c + D)$ bid by bid, so Π_θ^A equals the expected current profit inclusive of the continuation value of winning. This step does not use the construction cost parameters. Let $\mathbf{V}_\theta$ stack the state-specific value function $V_\theta(s, \bar{s})$ over the same grid. The stacked value function is

$$\mathbf{V}_\theta = (I - \beta M_\theta^0)^{-1} \mathbf{\Pi}_\theta^A. \tag{60}$$

The no-award matrix is required because $\mathbf{\Pi}_\theta^A$ already carries the value of prospective awards. I solve the linear system once per type on the empirical state support, then calculate the continuation gain of each observed award from the resulting $\mathbf{V}_\theta$ by applying the discount factor β. I then recover the construction cost parameters from the bid first-order condition.

Equivalent timing in the numerical implementation. The implementation stores values before the current exit draw. To relate these to $\mathbf{V}_\theta$, let S be the diagonal matrix with entries $1-\zeta(s,\bar{s})$, and let P_θ^0 be the no-award transition matrix before exit is applied. Under the conditional-value convention in (22),

$$M_\theta^0 = P_\theta^0 S, \qquad \widetilde{\mathbf{V}}_\theta = S\mathbf{V}_\theta.$$

The pre-exit values stored by the implementation therefore satisfy

$$\widetilde{\mathbf{V}}_\theta = (I - \beta S P_\theta^0)^{-1} S \mathbf{\Pi}_\theta^A.$$

For a focal award, let $\widetilde{Q}_\theta^a$ be its transition row before next-period exit, where $a \in \{\text{win}, \text{lose}\}$. The rows in (47) are $Q_\theta^a = \widetilde{Q}_\theta^a S$, so

$$\beta(Q_\theta^{\text{win}} - Q_\theta^{\text{lose}})\mathbf{V}_\theta = \beta(\widetilde{Q}_\theta^{\text{win}} - \widetilde{Q}_\theta^{\text{lose}})\widetilde{\mathbf{V}}_\theta.$$

Thus, incorporating exit into the next-period transition is an equivalent way to write the same continuation gains and bid inversion.

E Additional Estimates and Model Fit

This appendix reports additional first-stage and cost-equation estimates of the estimated specification to the ones reported in the main text. Appendix coefficients point estimates, with 95% percentile intervals from 300 firm-bootstrap draws. Table 2 in the main text reports the cost equation's type, state, scale, category and sector coefficients. Table 16 below reports the project controls.

E.1 Model fit

Table 17 compares selected statistics from the model's predictions compared with the data at observed states with qualified bidders. Overall, the model's predictions are close to the data. The model slightly over-predicts mean quarterly completion share (0.280 against 0.260) and the median submitted bid. It slightly under-predicts participation and the median winning bid.

Table 14: Quarterly completion share: ZOIB specification

Parameter	Estimate	95% CI
Zero-completion intercept, α_0^ξ	0.077	[−0.156, 0.290]
Zero-completion standardized log backlog, β_0^ξ	−0.001	[−0.188, 0.173]
Full-completion intercept, α_1^ξ	−1.851	[−2.114, −1.610]
Full-completion standardized log backlog, β_1^ξ	−1.331	[−1.619, −1.029]
Partial-mean intercept, α_μ^ξ	−0.480	[−0.639, −0.327]
Partial-mean standardized log backlog, β_μ^ξ	−0.806	[−0.942, −0.663]
Log type loading, $\log \lambda^\xi$	−0.392	[−1.252, −0.027]
Log Beta precision, $\log \phi$	0.738	[0.625, 0.891]
Logit H share, α^ω	−1.396	[−1.865, −0.971]

Notes: The nine-parameter finite-mixture completion model of Appendix C.2. The single type loading $\lambda^\xi = \exp(\log \lambda^\xi)$ shifts all three components toward faster completion for the High type. The backlog variable is standardized as in (32). At the median post-award backlog the implied expected quarterly completion share is 0.224 for the Low type and 0.423 for the High type. Confidence intervals are 95% percentile intervals from the 300 firm bootstrap draws.

Table 15: Participation logit

Parameter	Estimate	95% CI
Constant, δ_A	−3.450	[−4.083, −2.624]
Current backlog, δ_B	−0.242	[−0.339, −0.163]
Paid experience, δ_P	−0.488	[−0.650, −0.337]
Same category, δ_M	−0.728	[−1.030, −0.361]
Category gap, δ_R	−1.204	[−1.431, −0.947]
Mean backlog, $\delta_{\bar{B}}$	0.454	[0.285, 0.691]
Mean paid experience, $\delta_{\bar{P}}$	0.617	[0.349, 0.911]
Mean same category, $\delta_{\bar{M}}$	0.086	[−0.753, 0.729]
Mean gap, $\delta_{\bar{R}}$	0.870	[0.373, 1.301]
High-type intercept, δ_H	1.331	[1.054, 1.593]

Notes: A positive outcome is a submitted bid that is not disqualified before evaluation. Aggregate-state controls are estimated but omitted here. Confidence intervals are 95% percentile intervals from the 300 firm bootstrap draws.

Table 16: Additional second-stage coefficients: project costs

Parameter	Estimate	95% CI
Project controls		
BOQ log item count (standardized)	0.109	[0.005, 0.196]
BOQ item count missing	−0.163	[−0.824, 0.406]
Log advertised duration (standardized)	−0.017	[−0.093, 0.068]
Advertised duration missing	0.034	[−0.967, 1.237]
Procuring entity: district	−0.128	[−0.457, 0.249]
Procuring entity: government agency	−0.068	[−0.409, 0.285]
Procuring entity: hospital	0.009	[−0.338, 0.417]
Procuring entity: ministry	−0.019	[−1.211, 0.311]
Lagged material-price shock (standardized)	0.018	[−0.033, 0.060]
Material-price shock missing	−0.182	[−0.611, 0.420]

Notes: Coefficients are in units of the engineering cost estimate. The type, state, scale, firm-category, tender-category and sector coefficients of the same equation, and the cost dispersion σ, are reported in Table 2. Confidence intervals are 95% percentile intervals from the 300 firm bootstrap draws.

Table 17: Model fit statistics

Moment	Data	Model
Auction participation and bids		
Mean participating firms per tender-lot	2.13	1.89
Median bid / engineering cost estimate	0.981	1.023
Median winning bid / engineering cost estimate	0.923	0.800
Quarterly completion share		
Mean quarterly completion share	26.0%	28.0%

Notes: Simulated at observed states using the estimated participation and bid policies, on the 1,319 lots of the estimation sample; the winning bid is the lowest qualified bid, with the simulated rival standing in on one-qualified lots.

E.2 Counterfactual exhibits

Table 18: Opening tenders to lower categories

	Status quo	One category, smaller half	One category	Two categories
Projects awarded under both policies				
Spending (% change)	—	-2.21 (0.13)	-6.20 (0.17)	-9.63 (0.23)
Value-weighted delay (quarters change)	—	0.042 (0.007)	0.096 (0.011)	0.184 (0.015)
Competition, allocation, and award coverage				
Qualified bids per lot	1.65 (0.00)	2.10 (0.01)	2.44 (0.01)	2.49 (0.01)
High-type share of awarded ECE (pp change)	—	-3.46 (0.18)	-7.31 (0.27)	-13.82 (0.38)
Newly eligible firms' share of spending (%)	0.0	8.7 (0.1)	22.1 (0.3)	38.8 (0.4)
Projects receiving an award (%)	78.4 (0.1)	83.3 (0.1)	87.4 (0.1)	87.1 (0.1)
ECE receiving an award (%)	74.8 (0.4)	77.2 (0.4)	81.9 (0.4)	83.9 (0.4)

Notes: Auctions in categories E–A; 200 simulations per policy. The first expansion opens the smaller half of auctions in each category, defined by median ECE. Spending and delays compare projects awarded under both policies in each replication. Qualified bids and award coverage use all projects; the high-type share and newly eligible firms' share of spending use all awarded projects. Spending, delay, and high-type share changes are paired with the simulated status quo; the other rows report levels. Delays count unfinished work after the scheduled end, weighted by ECE and discounted at $\beta = 0.95$ per quarter. Simulation standard errors across 200 simulations are in parentheses, holding the estimated parameters fixed.

E.2.1 Delays when unawarded projects remain unfinished

I also compare the status quo with random promotion when projects receiving no award remain permanently unfinished. Unawarded projects remain fully unfinished and contribute to delays from the scheduled completion date onward. I average delays across all 1,319 projects using ECE weights and discounting at $\beta = 0.95$ per quarter. Across 200 matched simulations, value-weighted delay rises from 5.696 to 5.820 quarters, an increase of 0.125 quarters (simulation standard error 0.039), holding the estimated parameters fixed.

I also search for the promotion rule that minimizes this delay measure. I compare the same 1,000 rules using 50 matched simulations per rule, then evaluate the selected rule in 200 new matched simulations. Under this rule, value-weighted delay falls from 5.664 under its matching simulated status quo to 5.188 quarters, a reduction of 0.477 quarters (simulation standard error 0.042). Spending on projects awarded under both policies falls by 11.3 percent. Promotions rise from 111 to 298, and the share of ECE receiving an award rises from 75.3 to 78.7 percent. Delays among awarded projects fall only slightly, from 2.153 to 2.132 quarters. Most of the improvement therefore comes from reducing the value of projects left unawarded.